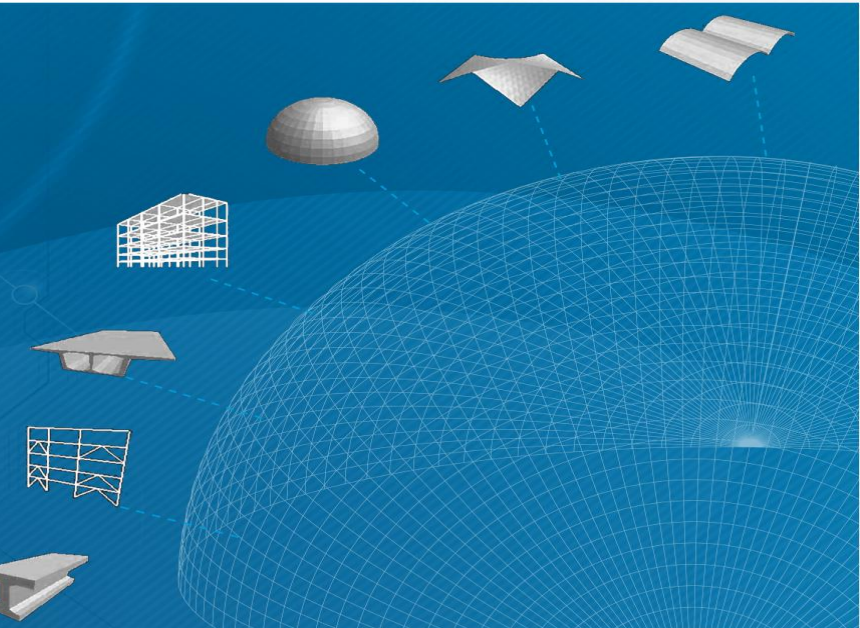
CE 412 Design of Concrete Structures

Lecture 5:

Cross-sections

Behavior in Combined Axial Load and Flexure – Columns



November – 2020

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Affiliate Faculty, Structural Engineering,
Asian Institute of Technology (AIT), Thailand

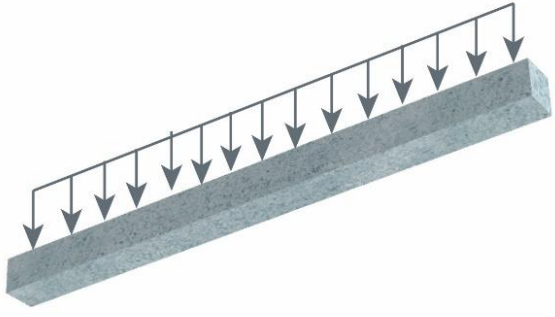
What is a Column?



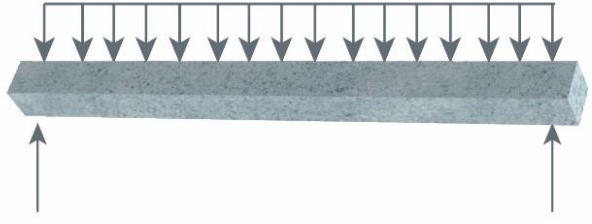
(a) Column



(b) Column or Beam



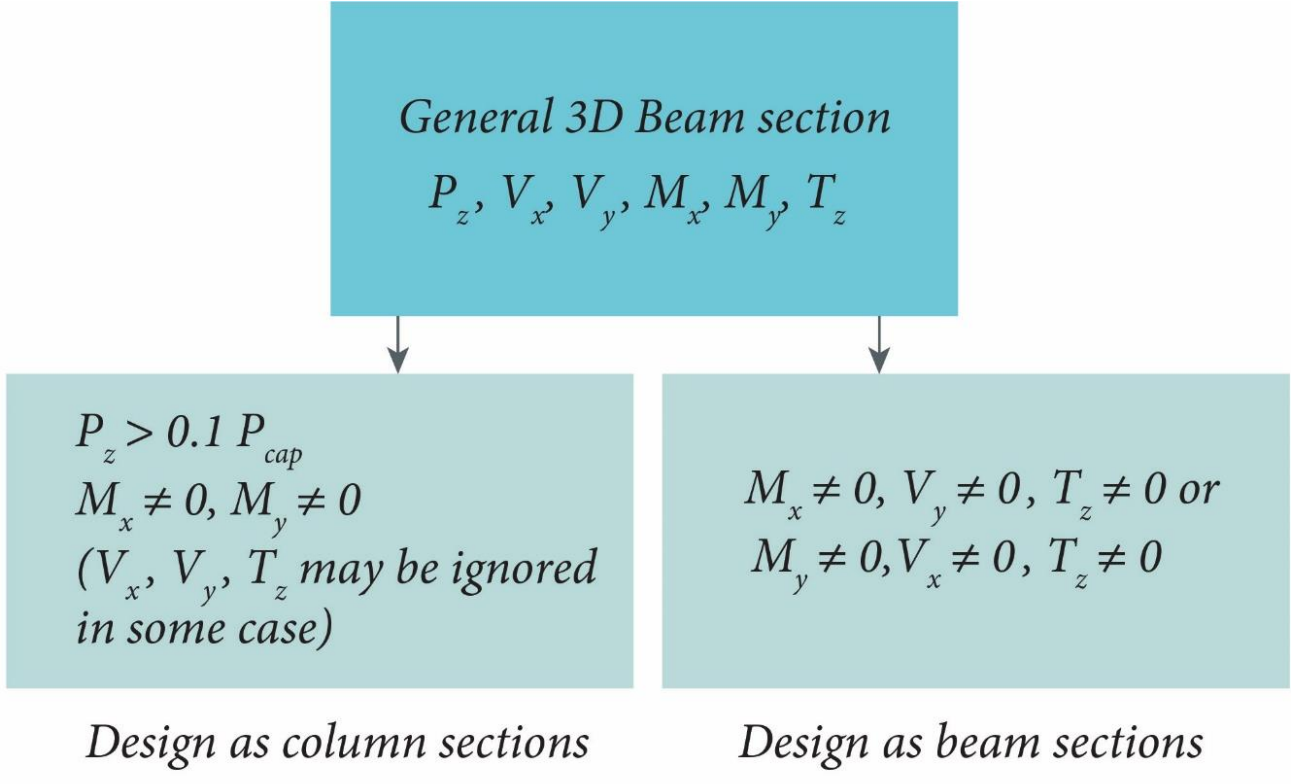
(c) Column or Beam



(d) Beam

What is a Column

- Generally no load between supports and are cast vertically.

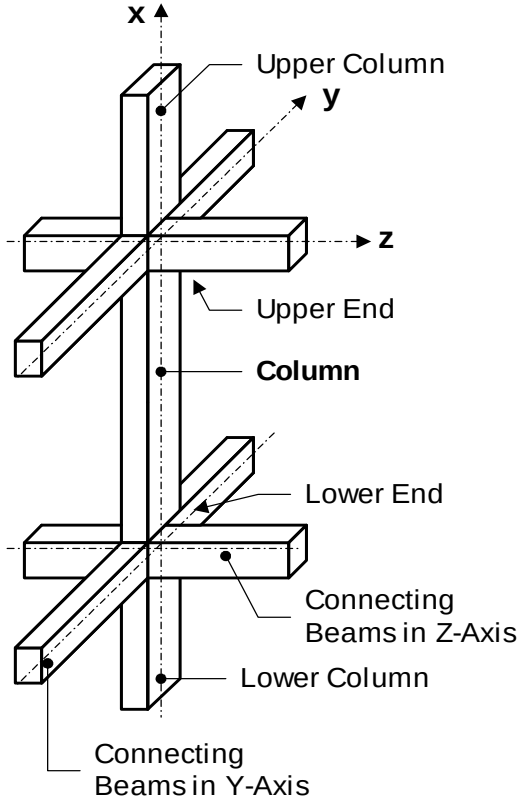


Classification of Columns

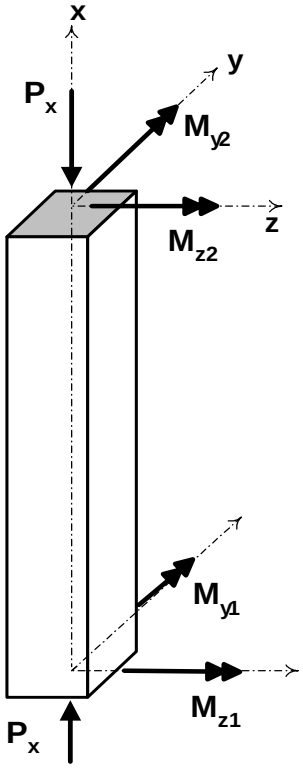
- Material
- Geometry
- Stability

The Column Design Problem: Given

- Loads:
  - Moments M_z, M_y, P_x at two ends
- Geometry:
  - Length, X-Section
  - Adjoining Members
- Material:
  - Concrete strength
  - Rebar Strength

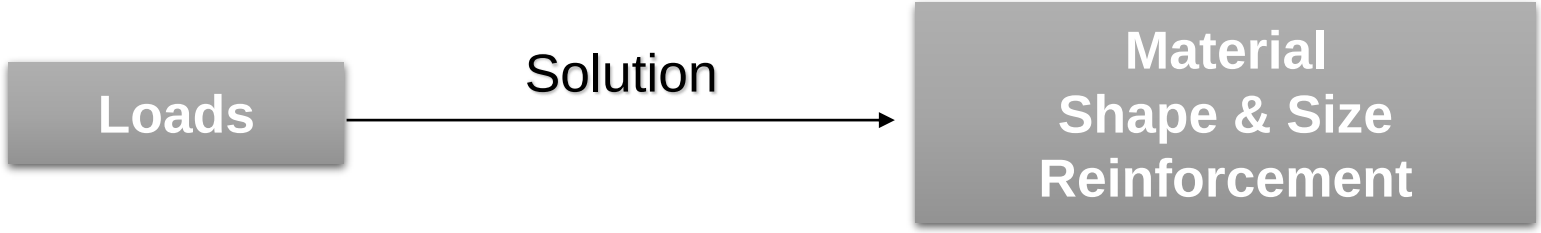


a) Basic Model

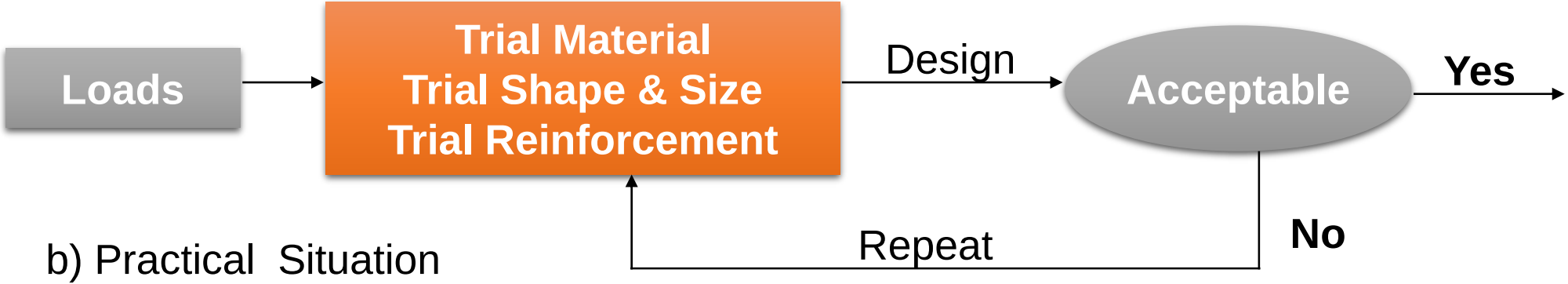


a) Column Loads

Column Design Problem



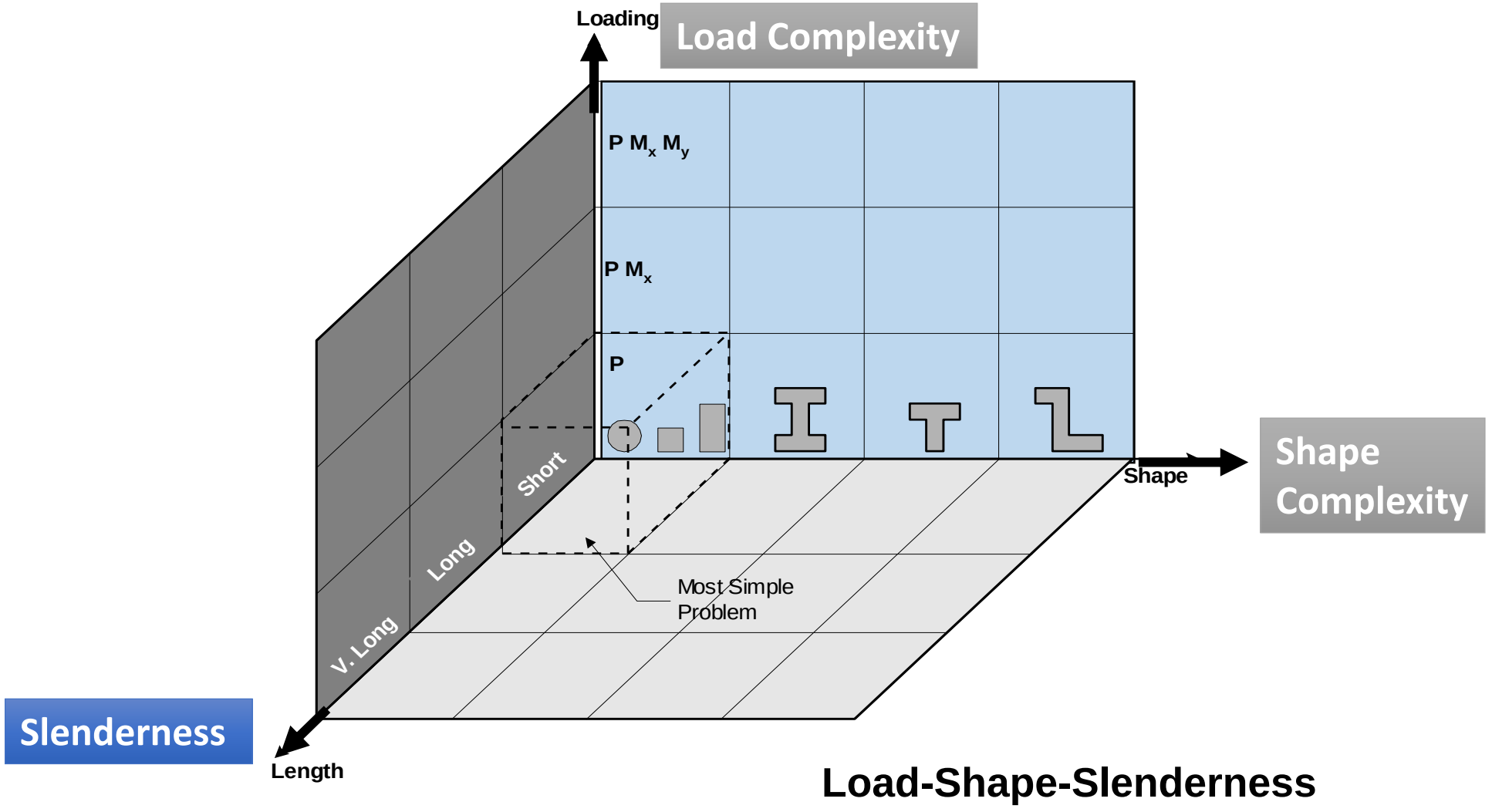
a) Ideal Situation



b) Practical Situation

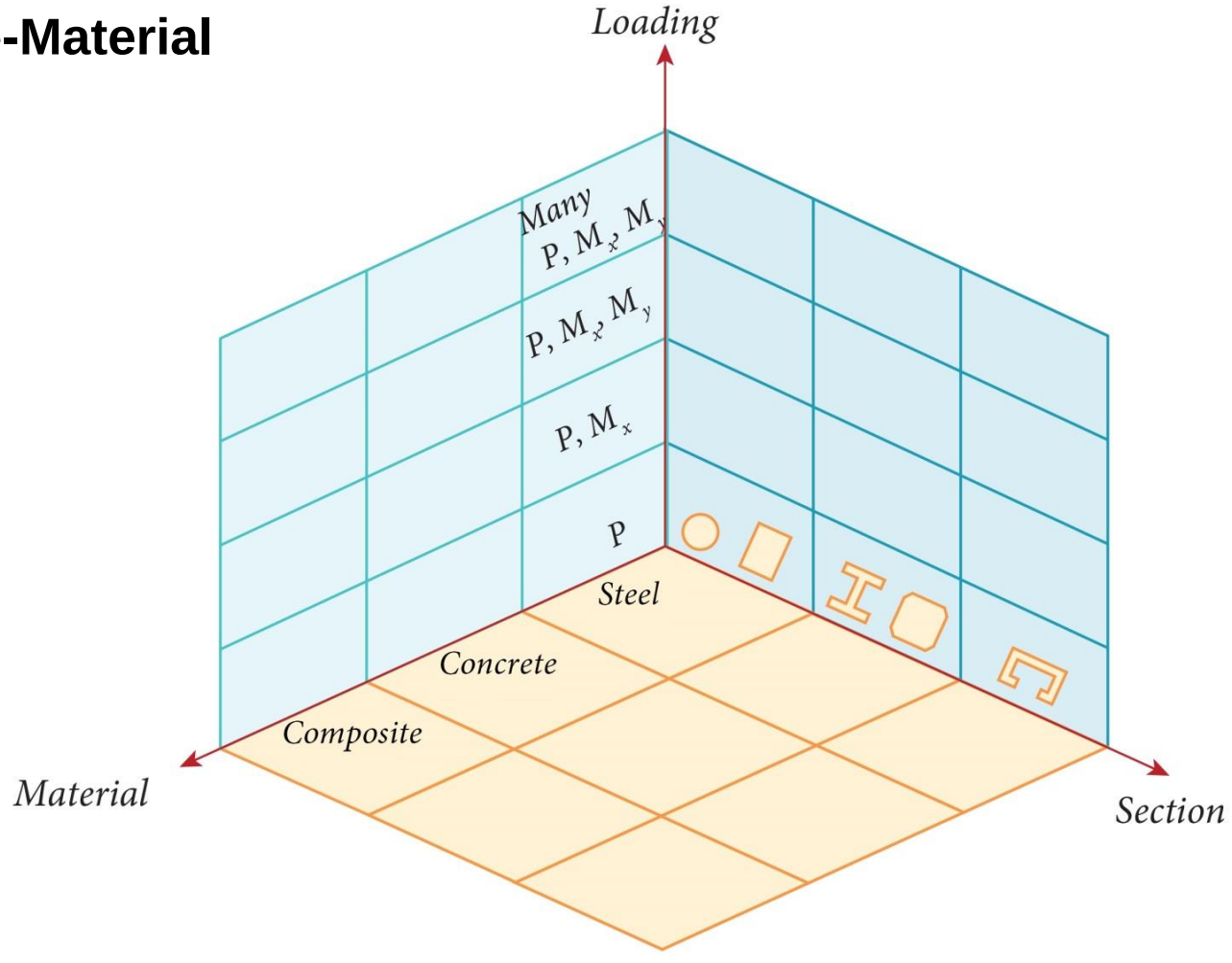
Complexity in the Column Design

- Loading
  - +P, -P, M_x, M_y
- Slenderness
  - Length (Short, Long, Very Long)
  - Bracing (Sway, Non-Sway, Braced, Unbraced)
  - Framing (Pin, Fixed, Free, Intermediate..)
- Section
  - Geometry (Rectangular, Circular, Complex..)
  - Materials (Steel, Concrete, Composite…)

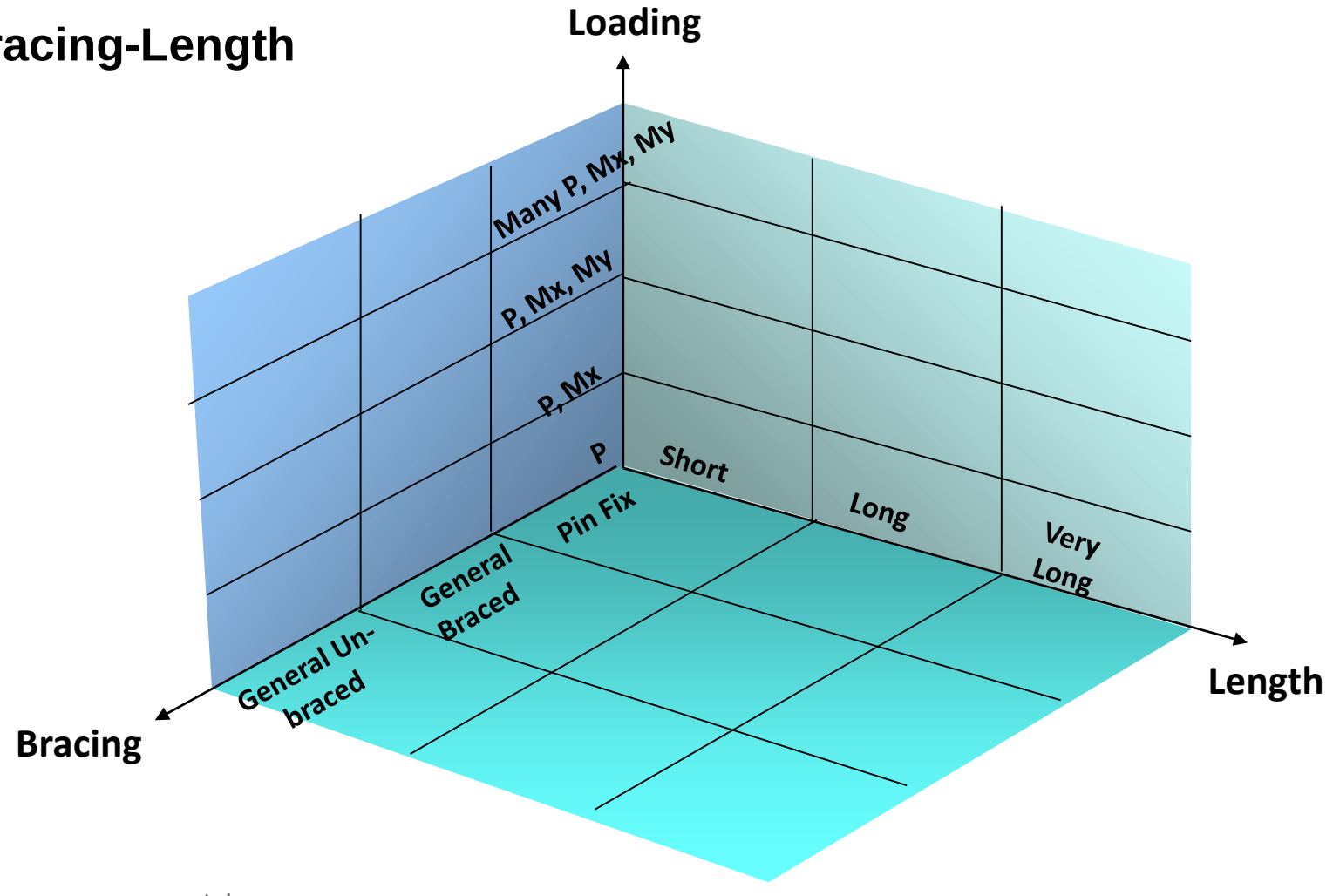


Load-Shape-Slenderness

Load-Shape-Material

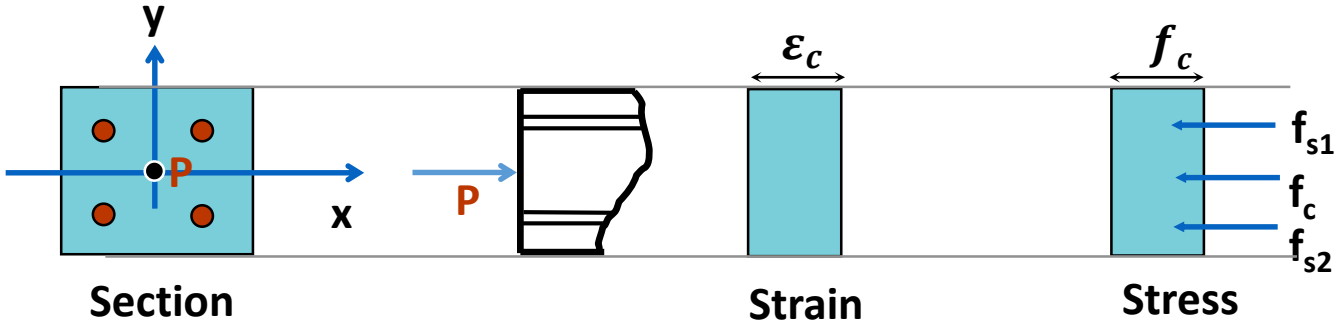


Load-Bracing-Length



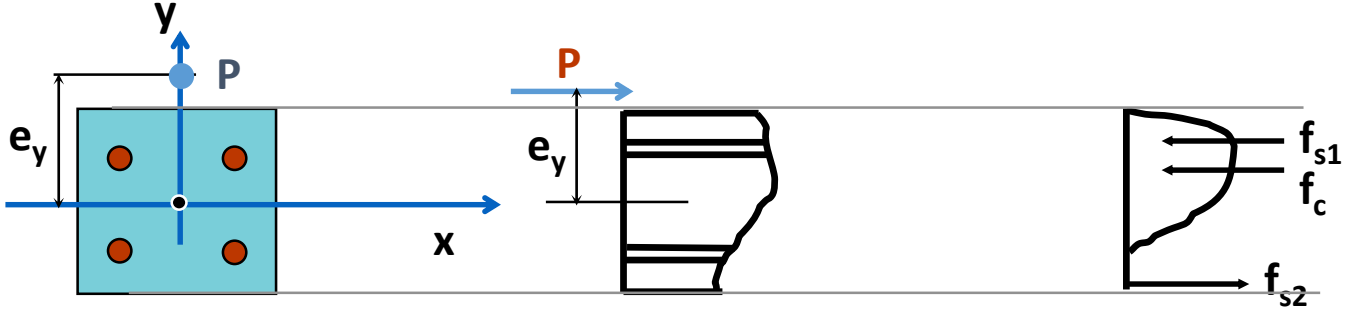
What is Uni-axial Bending

- Uni-axial bending is induced when column bending results in only one moment stress resultants about any of the mutually orthogonal axis.



No Bending

$M_x = 0, \quad M_y = 0$

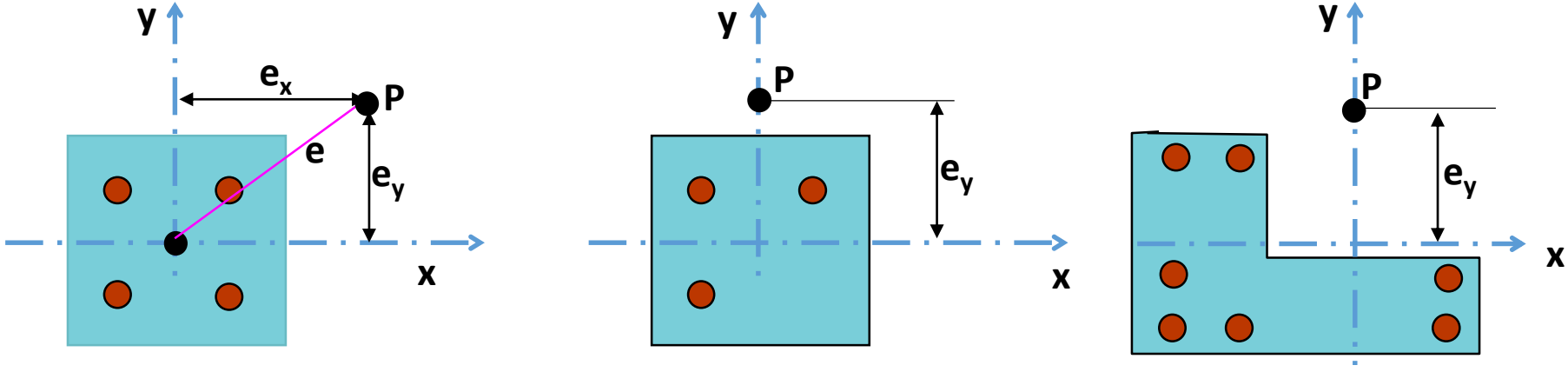


Uni-axial Bending

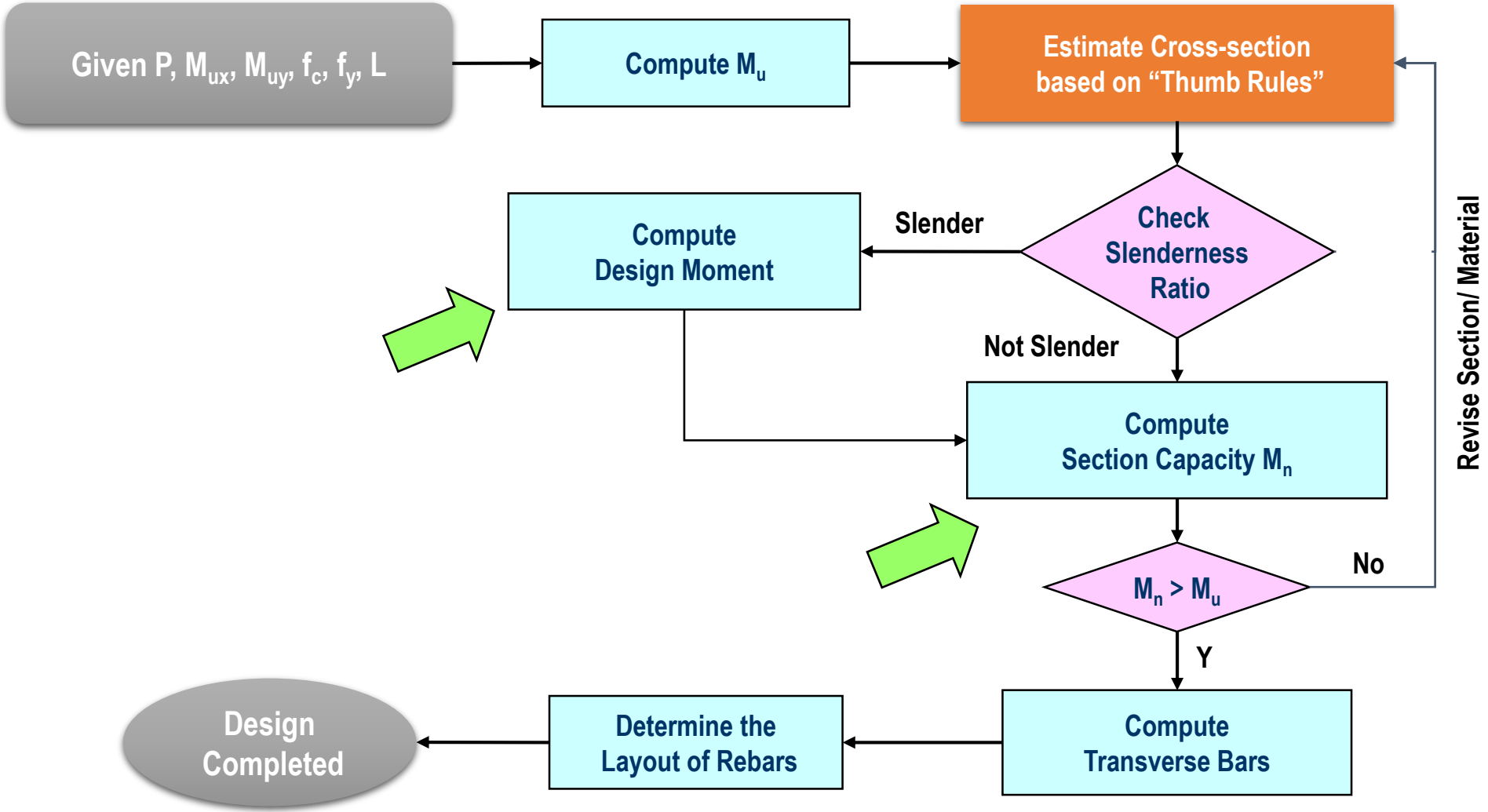
$M_x <> 0, \quad M_y = 0$

What is Bi-axial Bending

- Biaxial bending is induced when column bending results in two moment stress resultants about two mutually orthogonal axis.



Overall Design Process - ACI

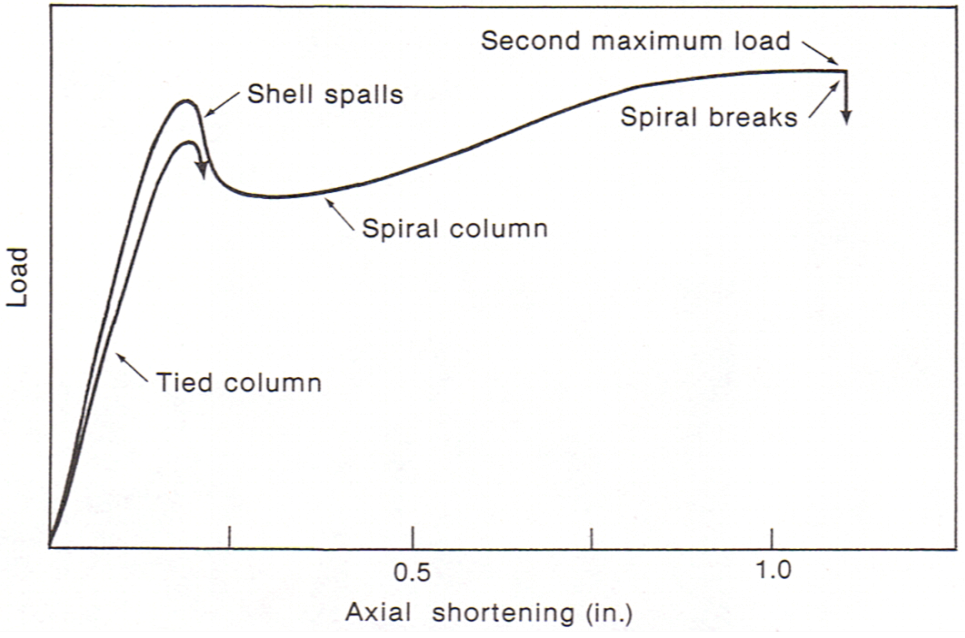


Main Design Steps

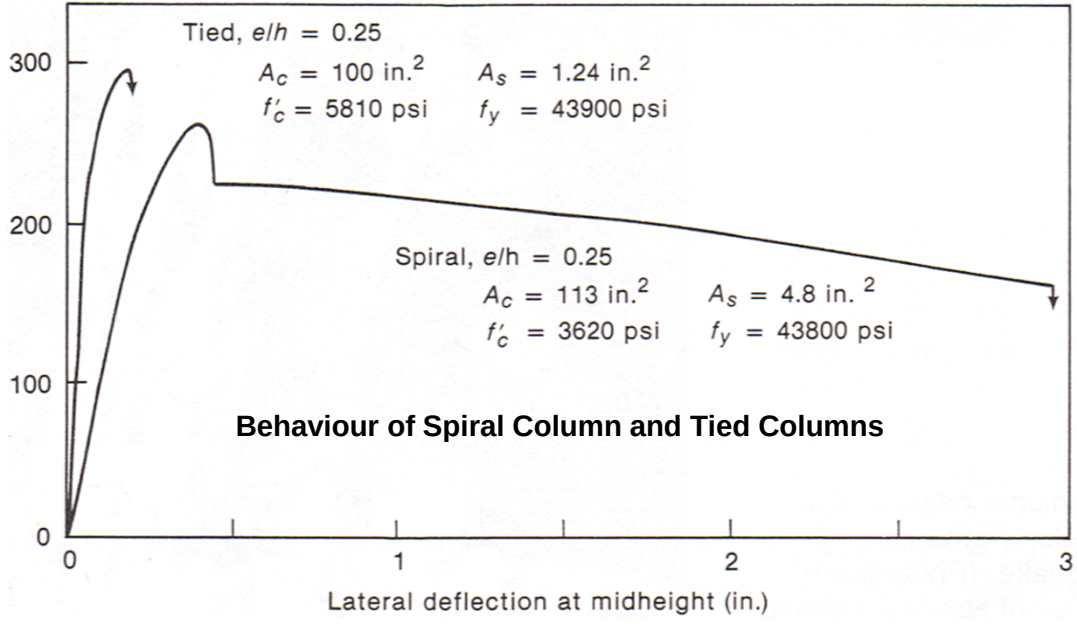
- Assume section dimensions
- Compute Design Actions
  - Elastic analysis results and magnification of moments due to slenderness, minimum eccentricities etc.
  - Direct determination of design actions using P-Delta or full nonlinear analysis
- Check Capacity for Design Actions
  - Assume failure criteria
  - Assume material layout and material models
  - Compute capacity and check against actions

Reinforced Concrete Short Columns

Axially loaded column



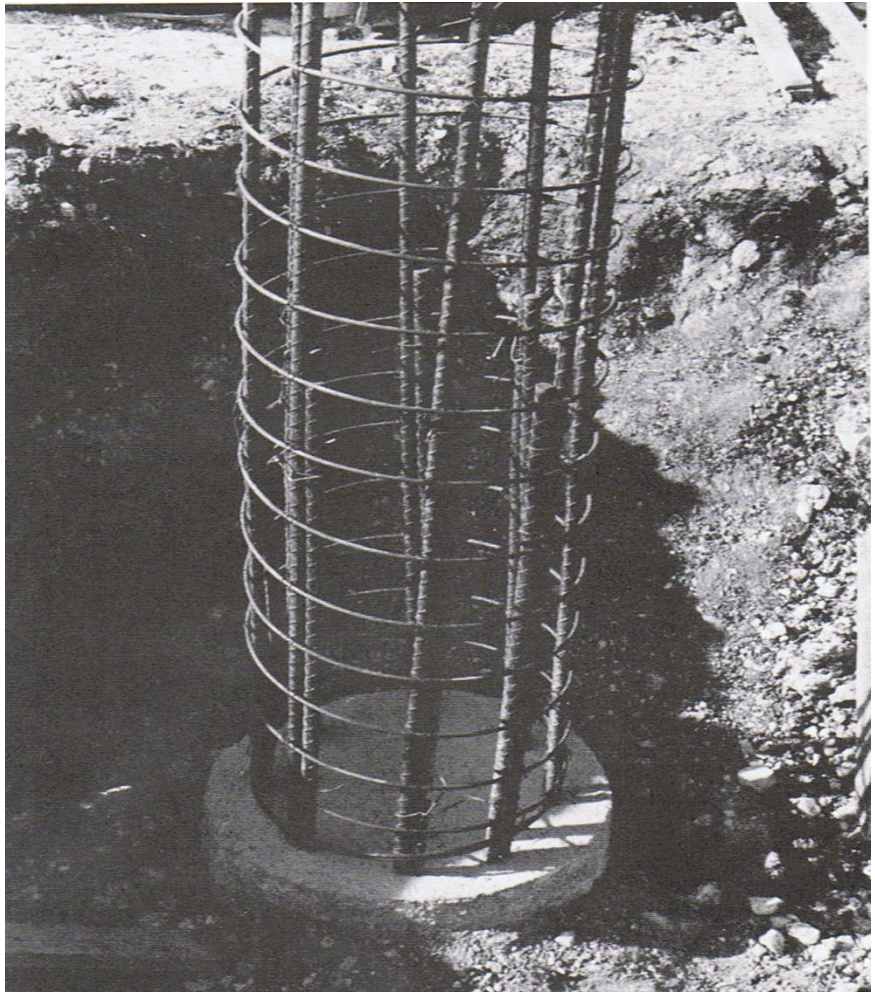
Eccentrically loaded column



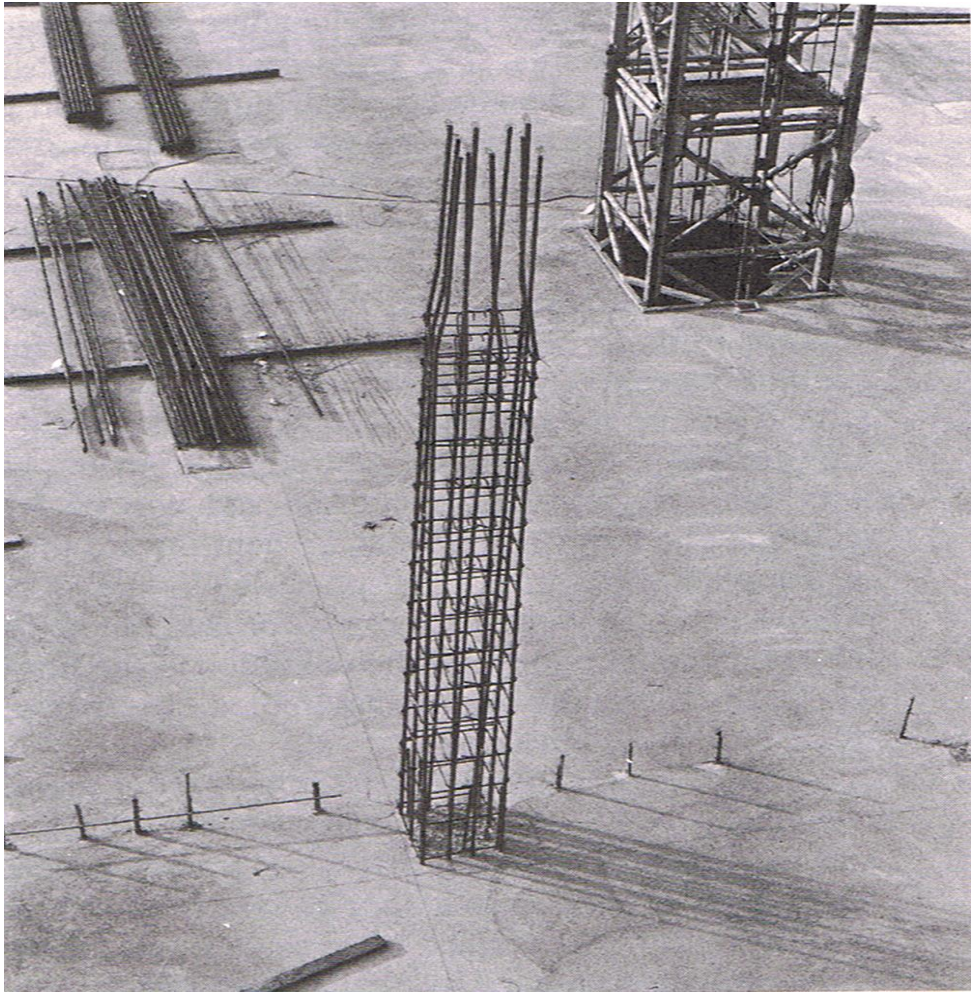
Behaviour of Spiral Column and Tied Columns

Reference: James G. MacGregor (1997) Reinforced Concrete Mechsnics and Design, Third Edition.

Spiral Column

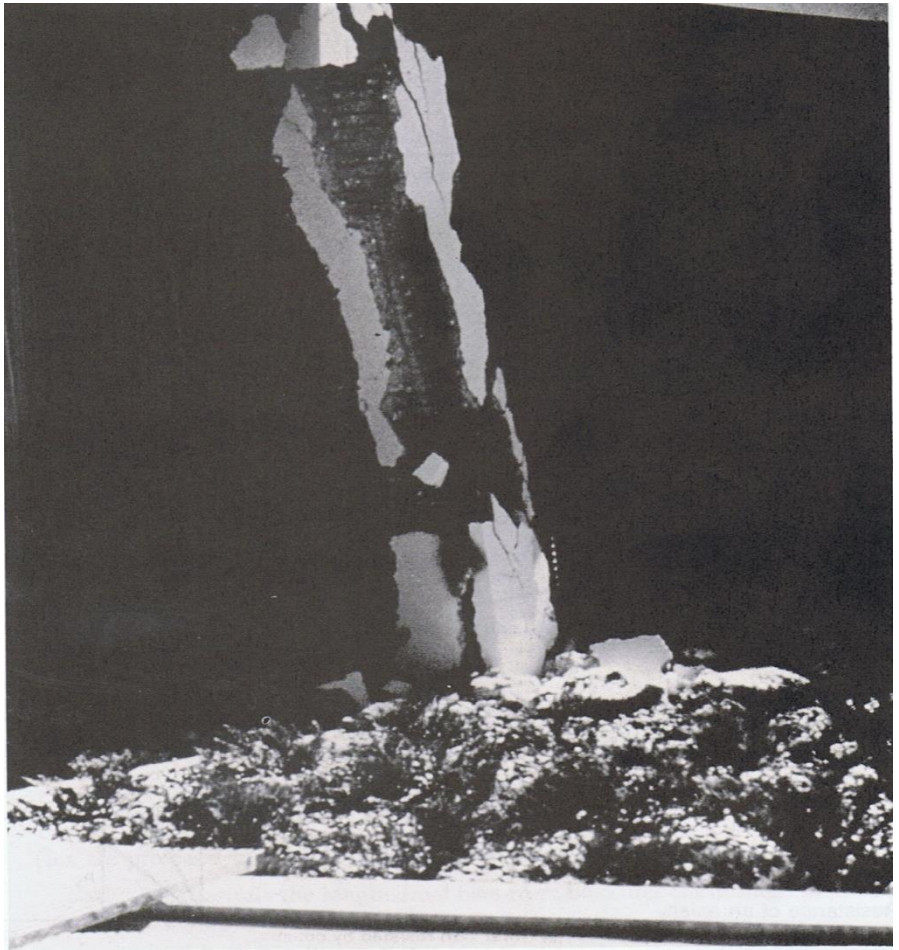


Tied Column



Reference: James G. MacGregor (1997) Reinforced Concrete Mechsnics and Design, Third Edition.

Spiral Column



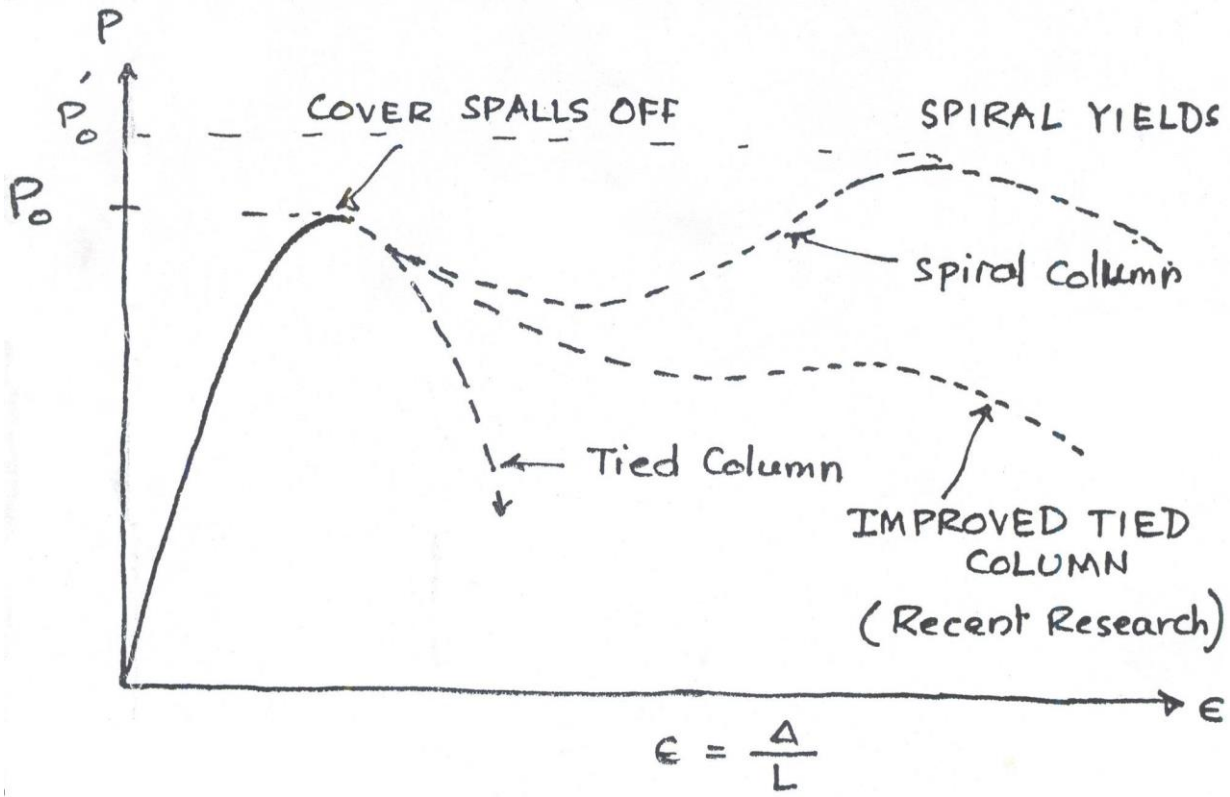
Tied Column



Damage by 1971 San Fernando Earthquake

Reference: James G. MacGregor (1997) Reinforced Concrete Mechsnics and Design, Third Edition.

Axially-Loaded Short Columns

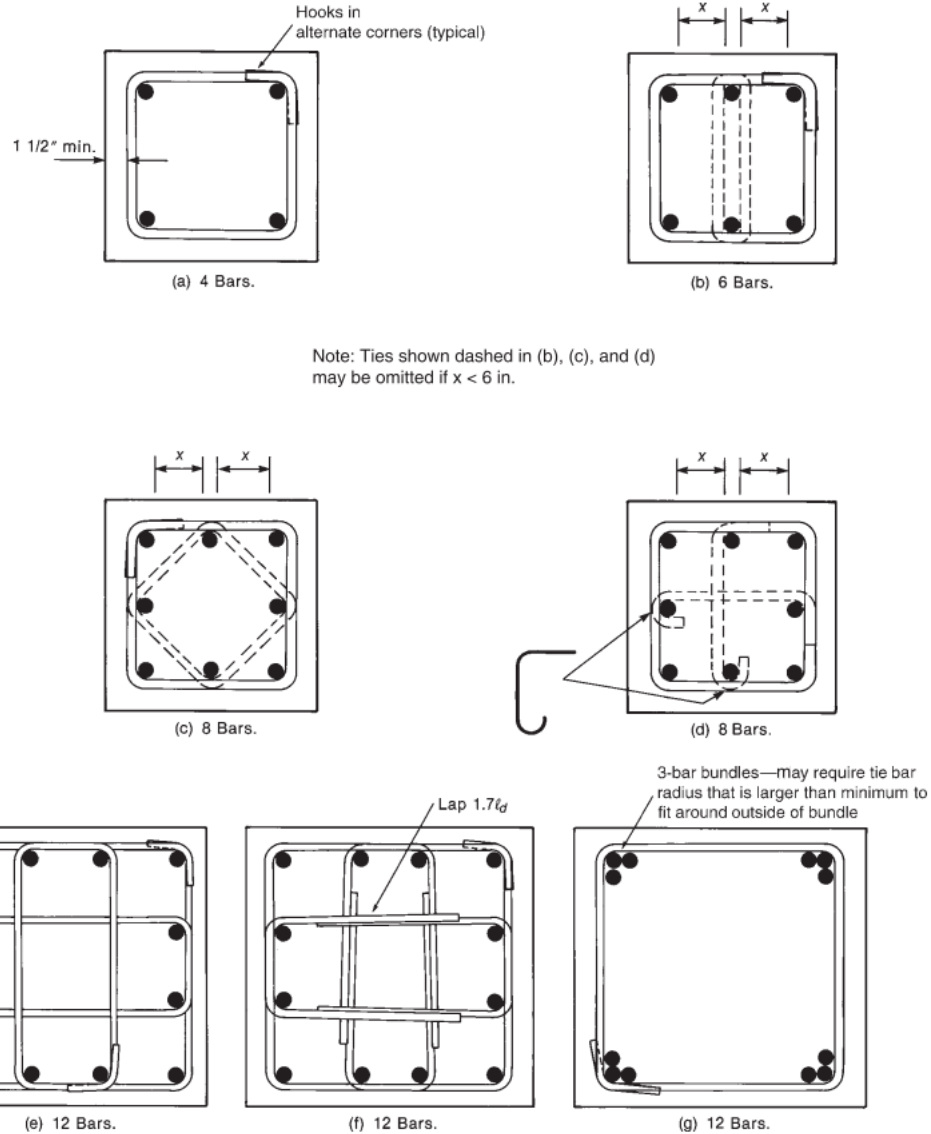


Minimum Spiral Reinforcement Ratio, ACI Section 10.9.3

$$\rho_s = 0.45\left(\frac{A_g}{A_c} - 1\right)\frac{f_c'}{f_{ys}} \qquad \text{(Eq. 10-5)}$$

- ACI Eq. 10-5 can be solved for max spacing.
- For spirals to bind the core, they must be spaced close together. ACI sec 7.10.4.3 limits the clear spacing to not more than 3 in.
- For concrete placing, spirals should be as far apart as possible. Clear spacing ≥ 1-⅓ times size of aggregate (ACI sec 3.3.3) and never less than 1 in (ACI sec 7.10.4.3) [also see ACI secs 7.9.1 & 7.10.4.6 to 7.10.4.8].

RC Columns: Basic Reinforcement Configurations and Tie Arrangements



Spacing and Construction Requirements for Ties

1. Ties restrain the longitudinal bars from buckling out through the surface of the column.
   - ACI sec 7.10.5.1

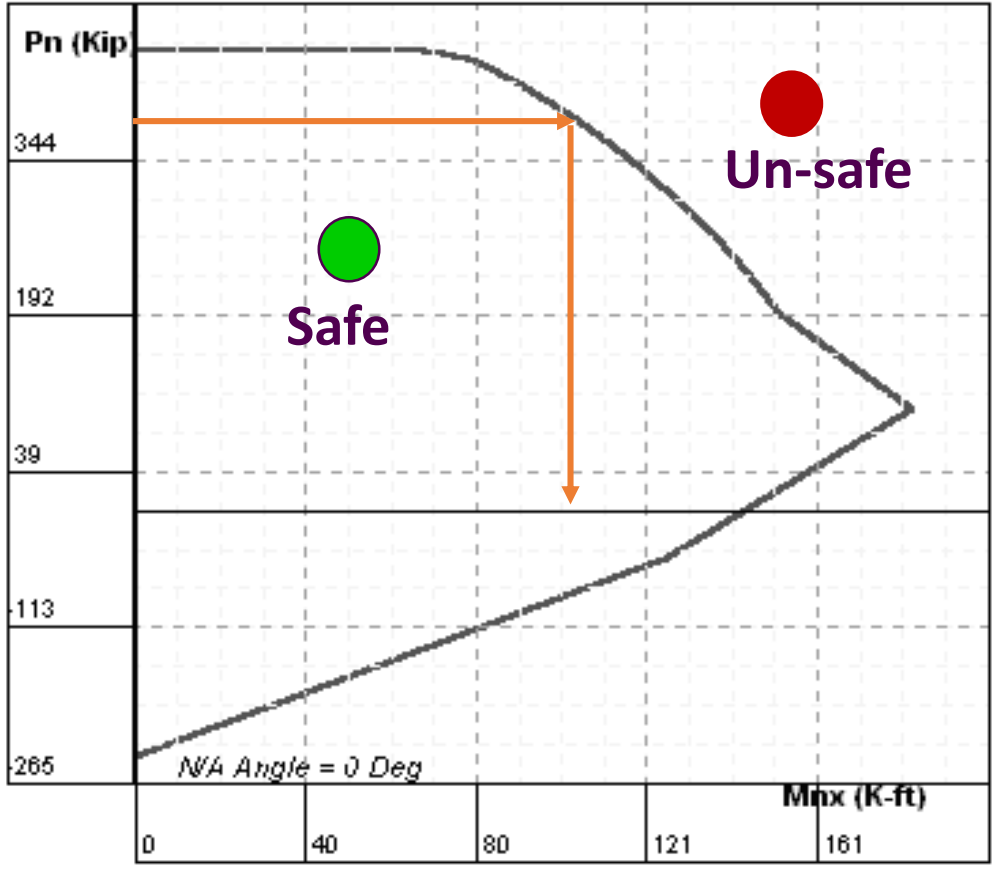
     # 3 bars for longitudinal # 10 bars
     # 4 bars for # 11, # 14, # 18 & bundled
   - ACI sec 7.10.5.2

     $s \leq$ 16 d_b longitudinal bars
     $\leq$ 48 d_b tie bars
     $\leq$ least dimension (b_{min})
   - ACI sec 7.10.5.3
2. Ties hold the reinforcement cage together during the construction process.
3. Properly detailed ties confine the concrete core, providing increased ductility.
4. Ties serve as shear reinforcement for columns
   - $s \leq d/2$ for effective shear reinforcement (ACI sec 11.5.4.1)
   - ACI eq 11-2 and 11-14, if shear governs.

Look for ACI sec 7.10.5.4, 7.10.5.5 and 7.9.1 for further requirements.

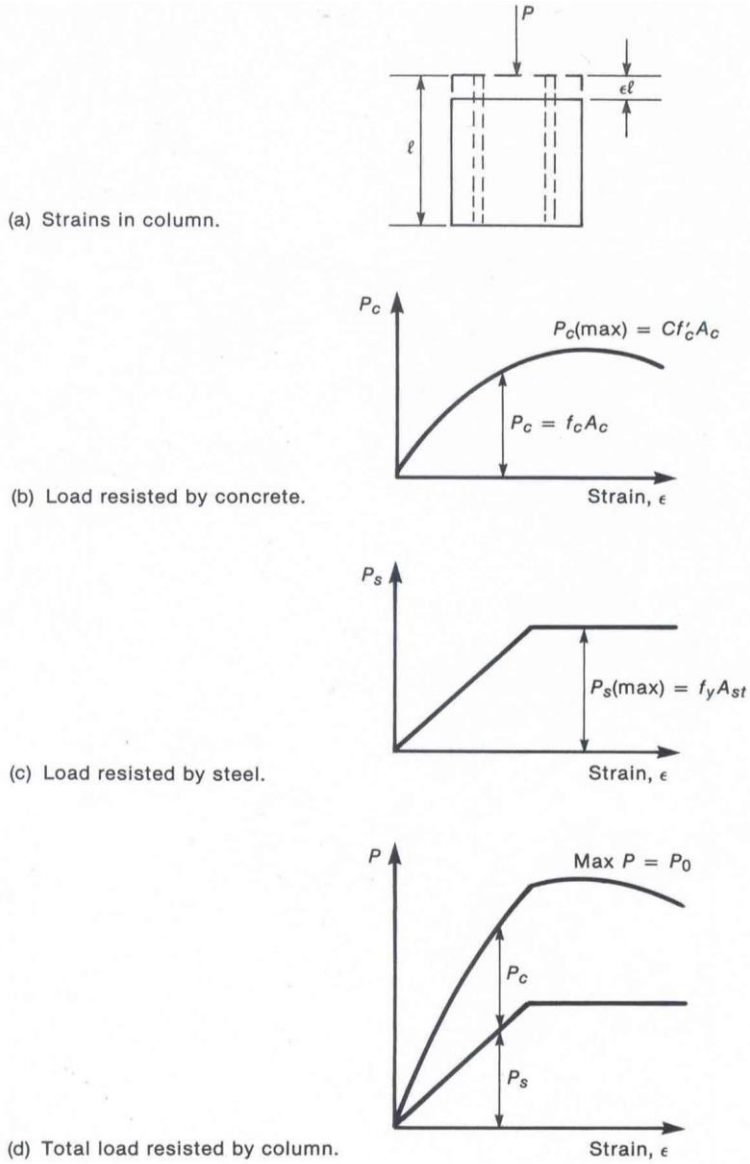
P-M Interaction Curve (Capacity Curve)

Uniaxial Bending - Short Columns



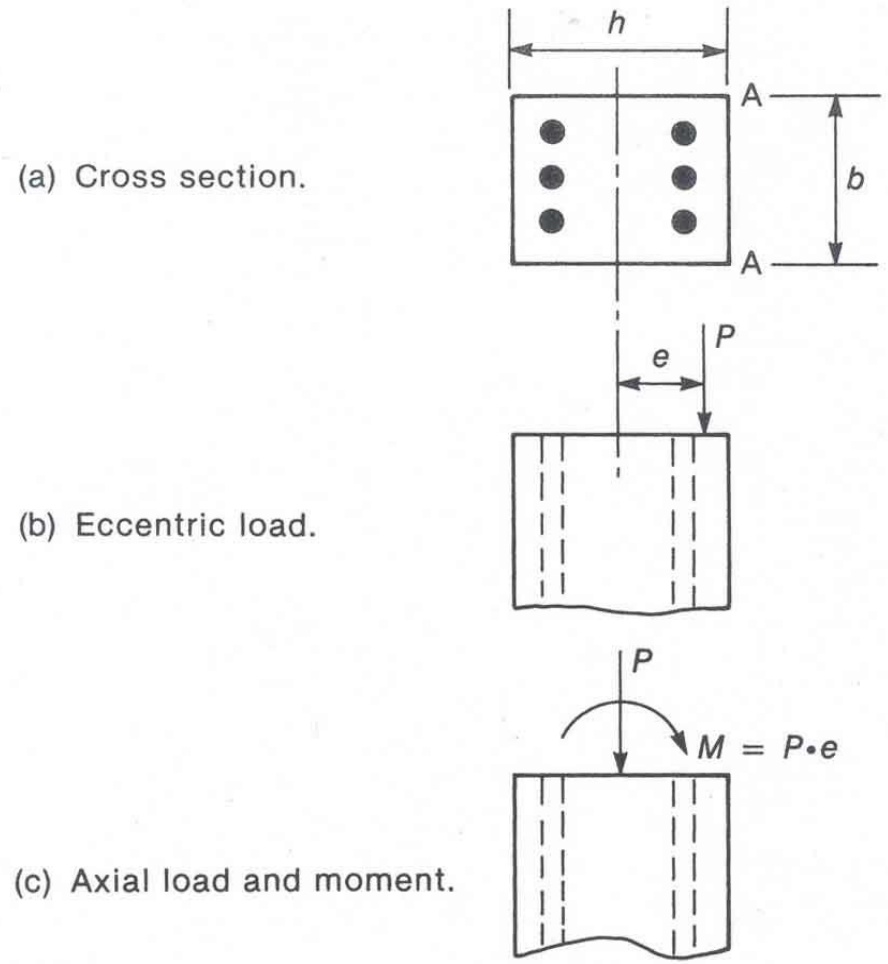
Capacity is property of the cross-section and does not depend on the applied actions or loads.

Column Capacity – Concentrically Loaded Columns



Column Capacity

Column Capacity – Eccentrically Loaded Columns



Column Capacity???

Column Capacity – Eccentrically Loaded Columns

$$\frac{P}{A} + \frac{My}{I} = f_{cu}$$

where

A, I = area and moment of inertia of the cross section, respectively
y = distance from the centroidal axis to the most highly compressed surface (surface A–A in Fig. 11-9a), positive to the right
P = axial load, positive in compression
M = moment, positive as shown in Fig. 11-9c

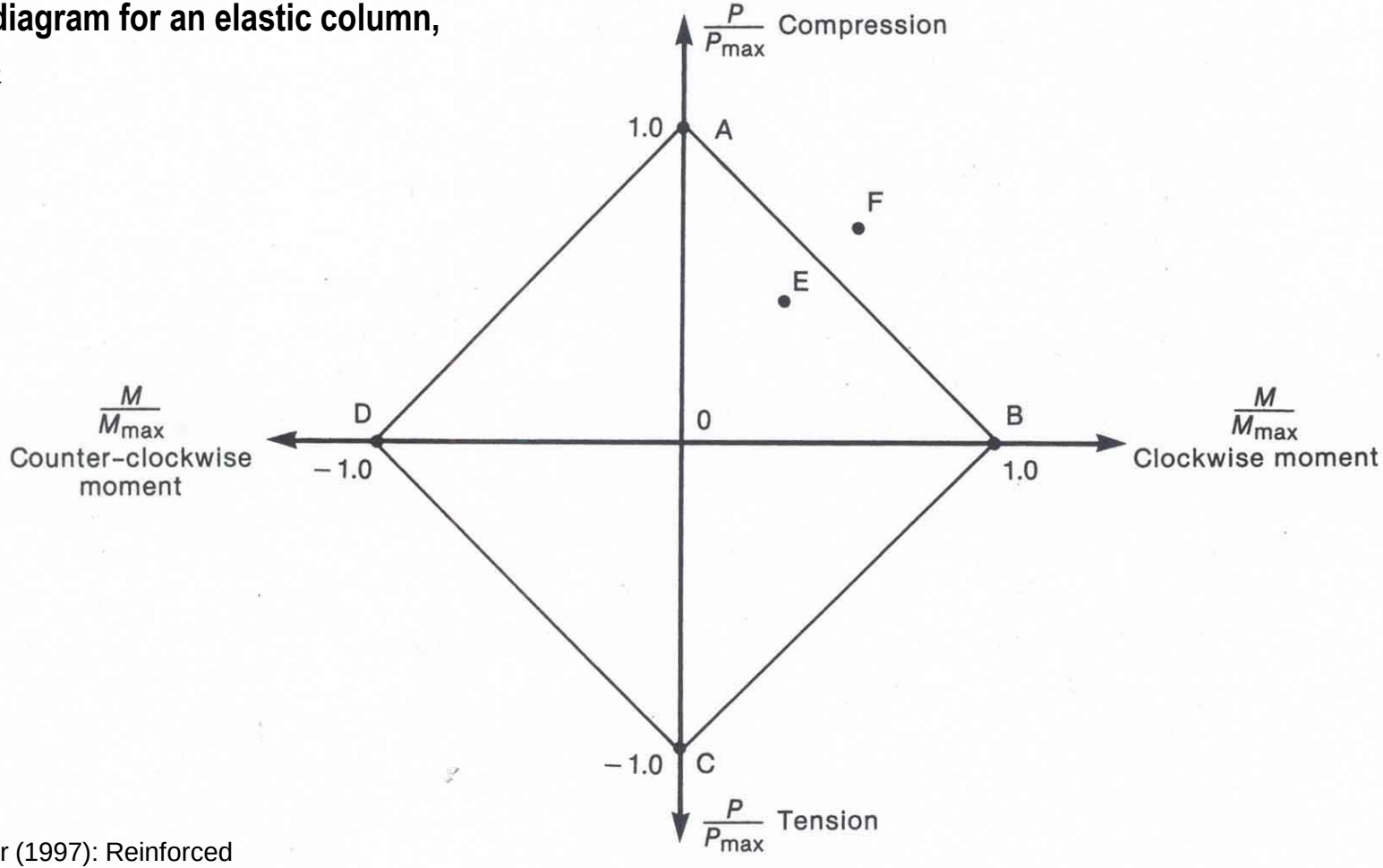
Dividing both sides of Eq. (11-2) by f_{cu} gives

$$\frac{P}{f_{cu}A} + \frac{My}{f_{cu}I} = 1$$

The maximum axial load the column can support occurs when $M = 0$ and is $P_{\max} = f_{cu}A$. Similarly, the maximum moment that can be supported occurs when $P = 0$ and M is $M_{\max} = (f_{cu}I/y)$. Substituting $P_{\max}$ and $M_{\max}$ gives

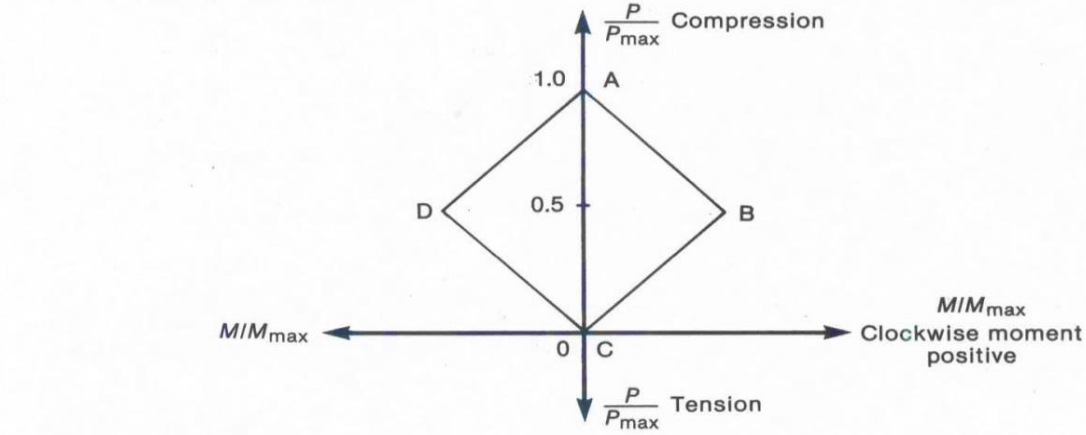
$$\frac{P}{P_{\max}} + \frac{M}{M_{\max}} = 1 \qquad (11\text{-}3)$$

Interaction diagram for an elastic column,
$f_{cu} = f_{tu}$

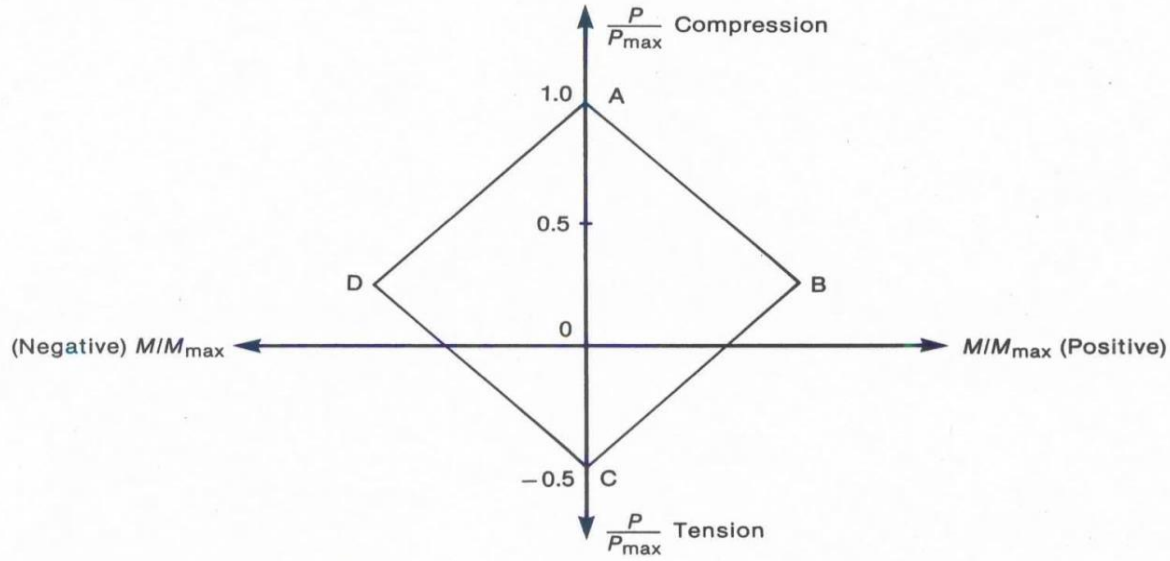


J. G. Macgregor (1997): Reinforced Concrete Mechanics and Design, 3rd Edition.

Effect of Compressive or Tensile Strength on Interaction curve



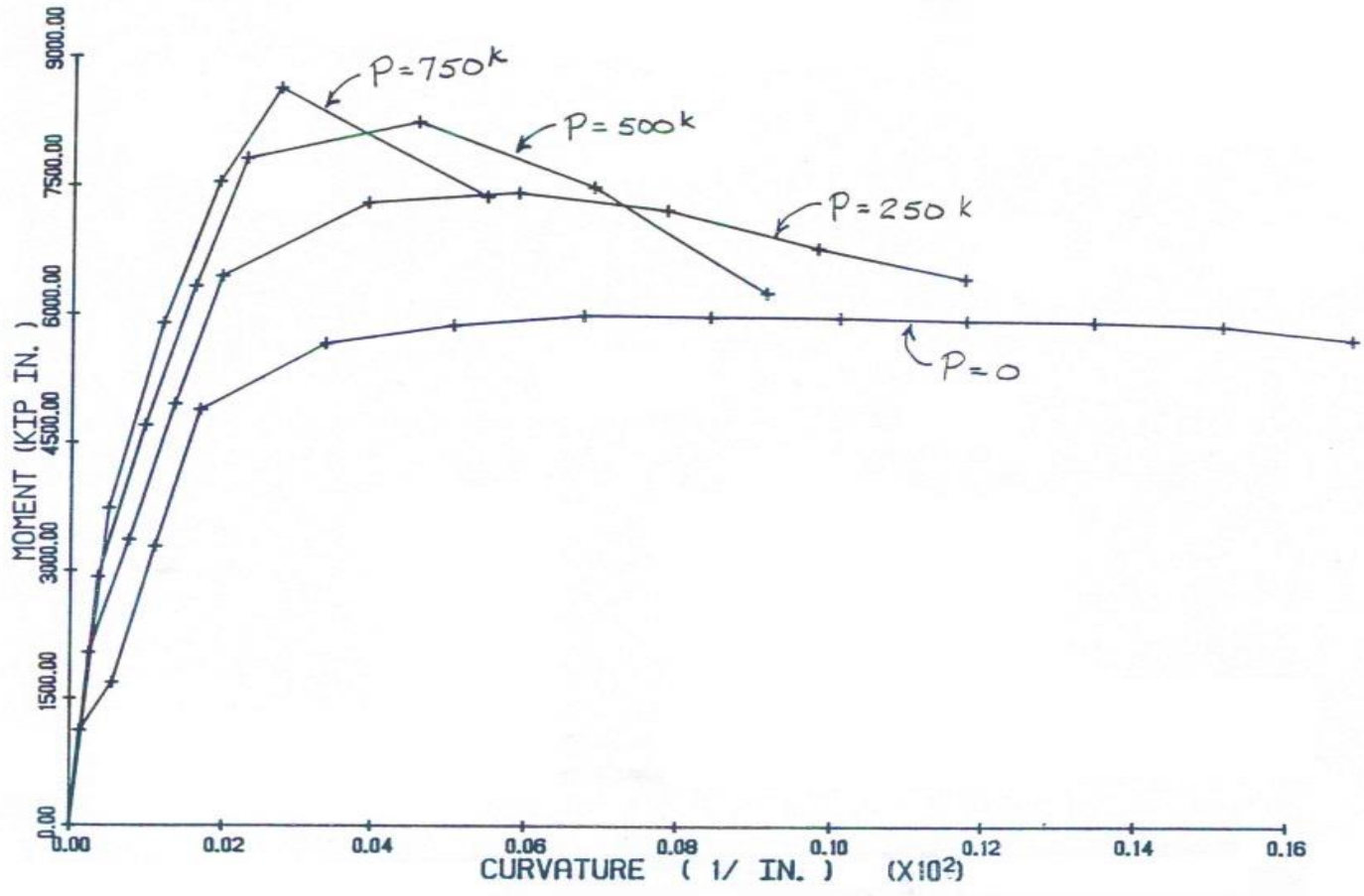
(a) Material with $f_{cu} = 0$.



(b) Material with $f_{tu} = -1/2f_{cu}$.

J. G. Macgregor (1997): Reinforced Concrete Mechanics and Design, 3rd Edition.

Effect of Axial Load on Moment-Curvature Curve



Uniaxial Bending

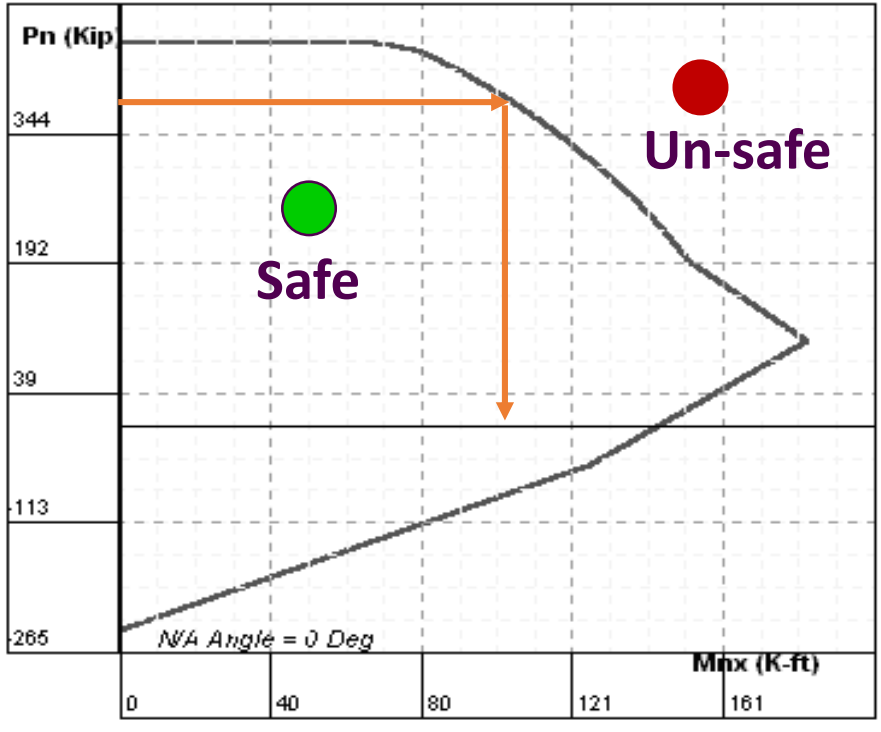
⇩

P-M Interaction Curve

$$N_z = \phi_1 \left[\frac{1}{\gamma_1} \int_x \int_y \sigma(x,y)\,dx\,dy \ \ldots + \frac{1}{\gamma_2} \sum_{i=1}^{n} A_i \sigma_i(x,y) \ \ldots \right]$$

$$M_x = \phi_2 \left[\frac{1}{\gamma_1} \int_x \int_y \sigma(x,y)\,dx\,dy\,.\,y \ \ldots + \frac{1}{\gamma_2} \sum_{i=1}^{n} A_i \sigma_i(x,y)\ y_i \ \ldots \right]$$

- The curve is generated by varying the neutral axis depth.



An Example P-M Interaction Diagram used for the Design of Short RC Columns

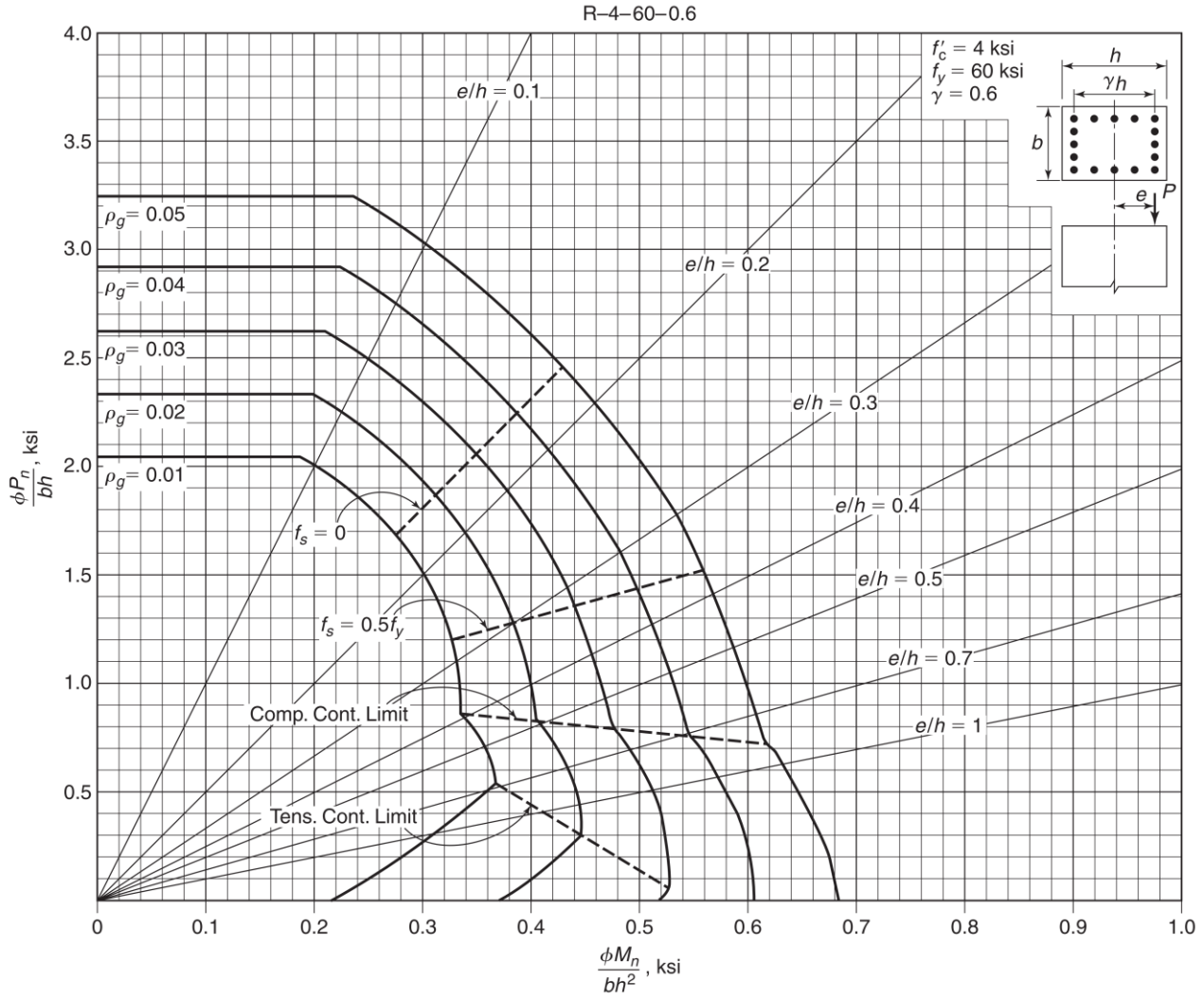


Fig. A-9a
Nondimensional interaction diagram for rectangular tied column with bars in four faces: $f'_c = 4000$ psi and $\gamma = 0.60$.

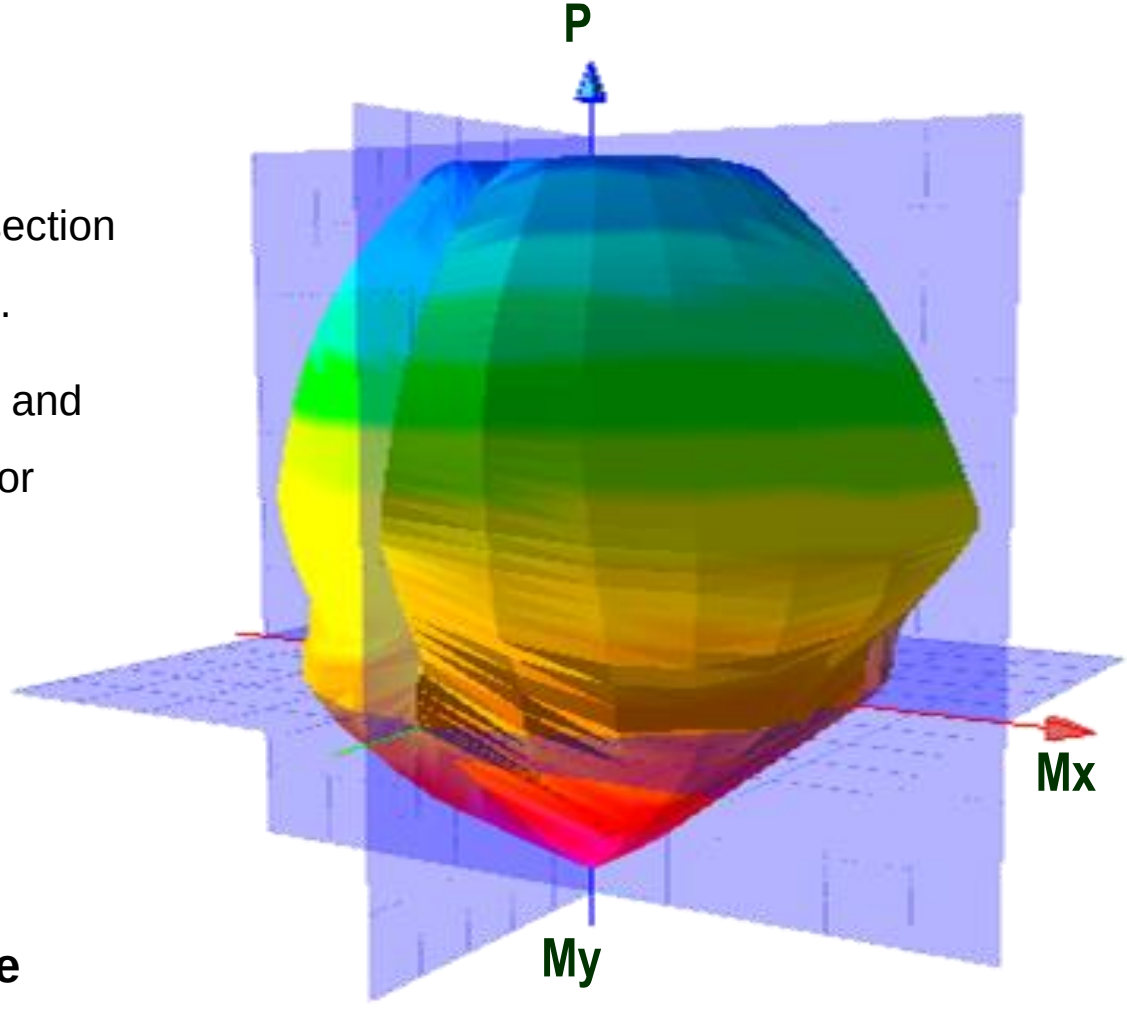
What is Capacity?

- The axial-flexural capacity of the cross-section is represented by three stress resultants.
- Capacity is property of the cross-section and does not depend on the applied actions or loads.

Biaxial Bending

⇩

P-M Interaction Surface



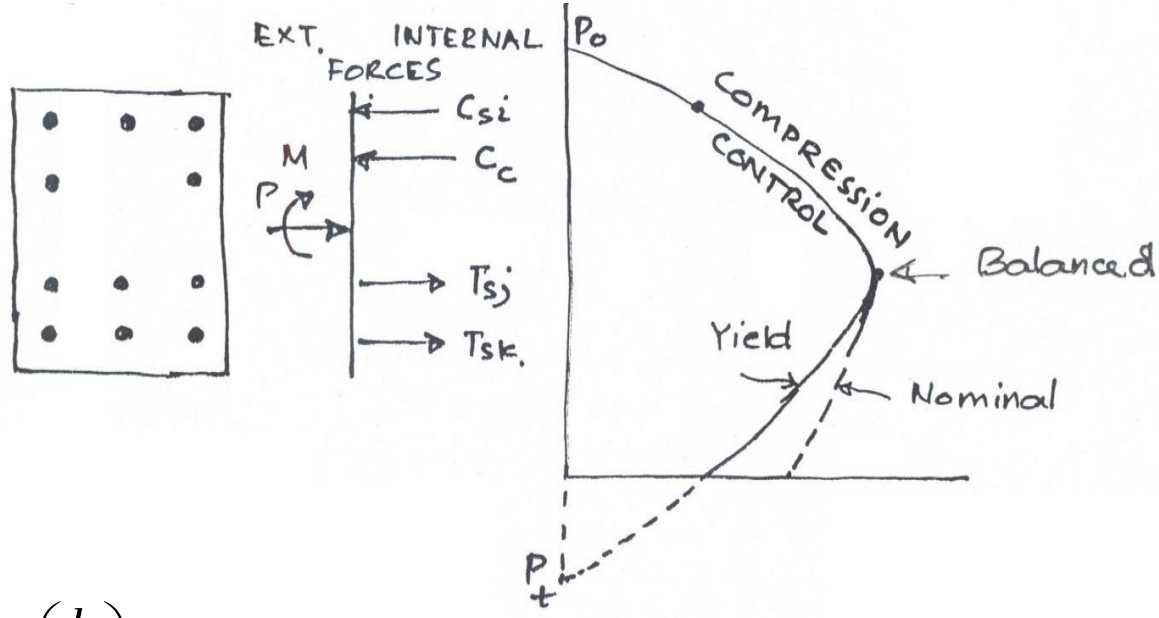
What is Capacity?

- Capacity is dependent on failure criteria, cross-section geometry and material properties.
  - *Maximum strain*
  - *Stress-strain curve*
  - *Section shape and Rebar arrangement etc.*

Development of P-M Interaction Diagrams for Uniaxially Loaded RC Short Columns – Solved Example

Courtesy: Dr. Khaliq ur Rashid Kayani (Architectural & Civil Engineering Services (ACES), Pakistan).

General Points on P-M Interaction Diagram



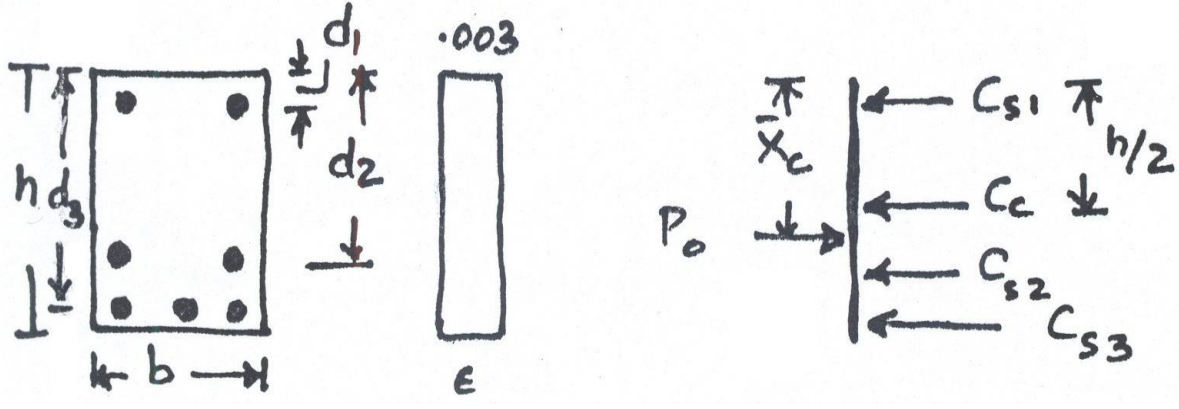
Axial Load (Compressive)

$$P_o \bar{X}_c = C_{si} d_1 + C_{s2} d_2 C_{s3} d_3 + C_c \left(\frac{h}{2} \right)$$

$$C_{si} = A_{si} \left(f_y - 0.85 f_c' \right)$$

$$C_c = 0.85 f_c' b.h$$

$$P_o = C_c + \sum_{i=1}^{3} C_{si}$$

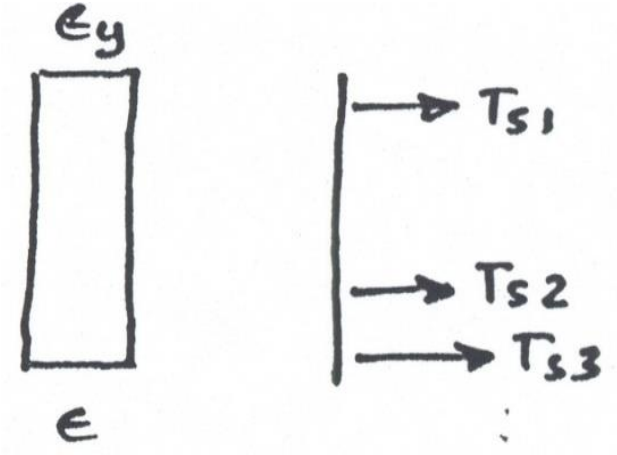


Axial Load (Tensile)

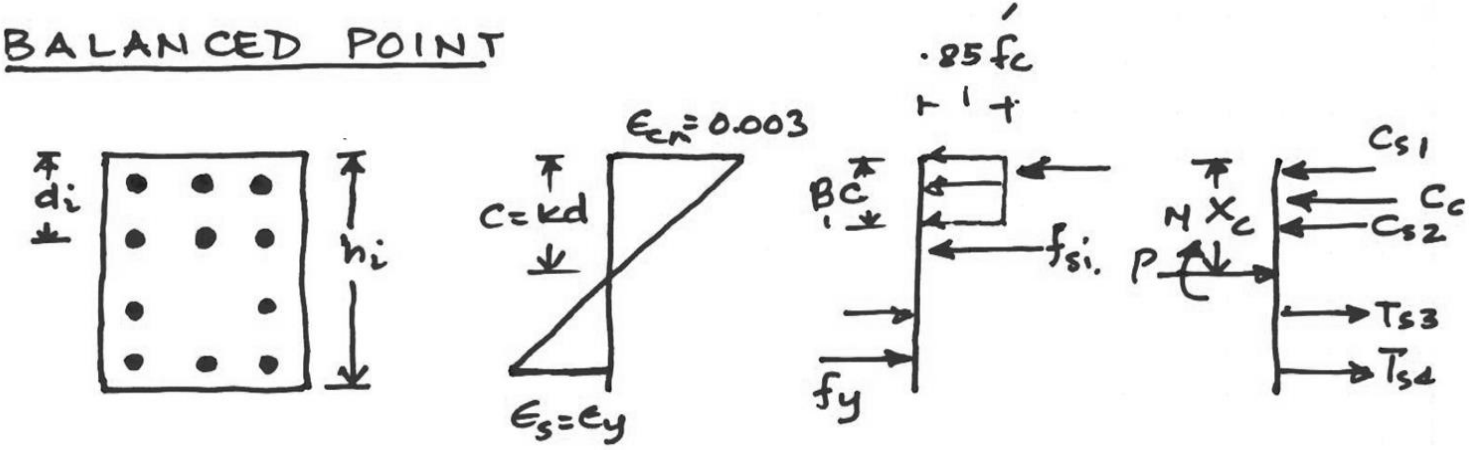
$$P_t = \sum_{i=1}^{3} T_{si}$$

$$T_{si} = A_{si}\ f_y$$

$$P_t \bar{X}_t = \sum_{i=1}^{3} T_{si}\,(d_i)$$



BALANCED POINT

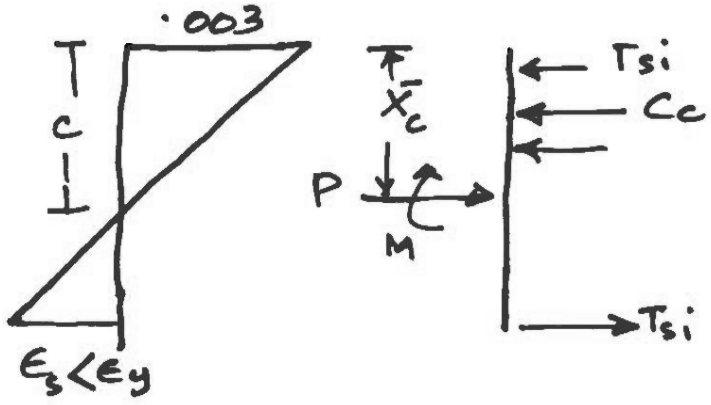


- $C_c = 0.85 f'_c \, b \, \beta_1 c$
- $C_{si} = A_{si} (f_{si} - 0.85 f'_c)$
  - Subtract $0.85 f'_c$ even if within $\beta_1 c$
  - Drop if -ve / stress
- $T_{si} = A_{si} f_{si}$
- $P = C_c + \Sigma C_{si} - \Sigma T_{si}$
- $\Sigma M (\text{at } \bar{x}_c)$

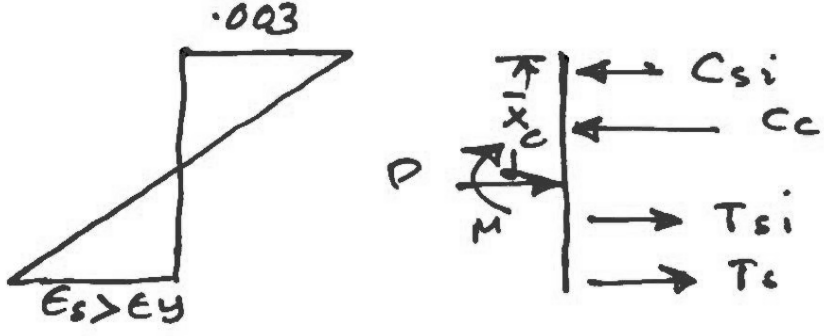
$$= C_c \left(\bar{x}_c - \frac{\beta_1 c}{2}\right) + \Sigma C_{si} (\bar{x}_c - d_i) + \Sigma T_{si} (d_i - \bar{x}_c)$$

(+ve n.a.)

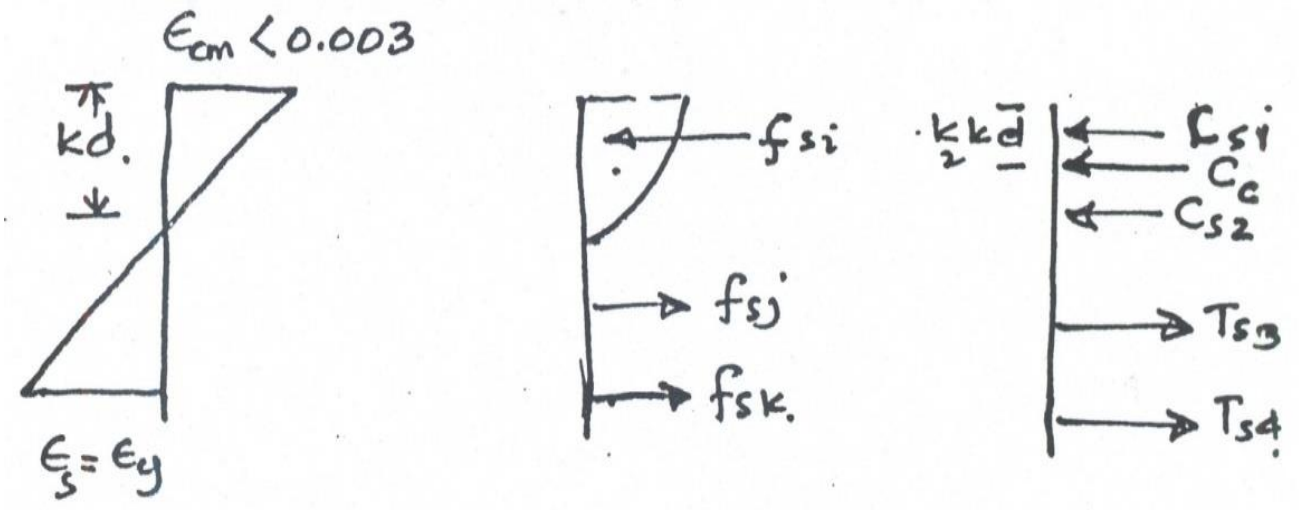
POINTS ABOVE BALANCED



POINTS BELOW BALACED - NOMINAL



Points below Balanced - Yield



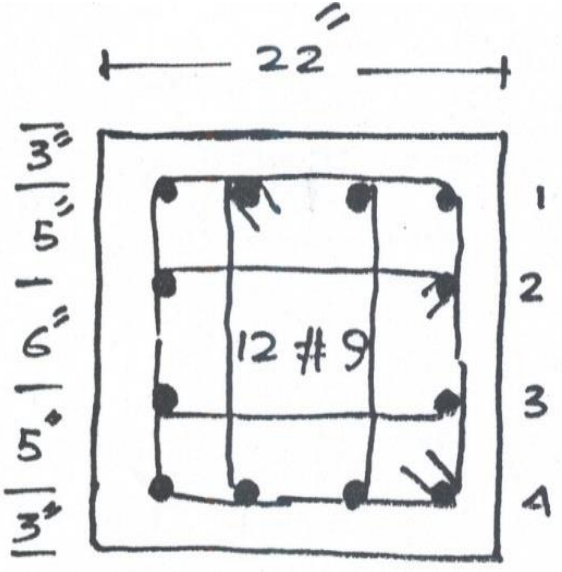
$$C_c = K_1 f_c' bkd$$

$$C_{si} = A_{si}\left(f_{si} - f_{ci}\right) \text{ should always be +ve}$$

Solved Example

Symmetric Section

$$\bar{X}_c = \bar{X}_t = \frac{h}{2} = 11''$$



$$f_y = 60\ ksi$$

$$\varepsilon_y = 0.00207$$

$$f_c' = 4000\ psi$$

Section dimensions					
Width =	22	Height =			
Steel areas and depths					
Layer no.	1	area =	4	depth =	3
Layer no.	2	area =	2	depth =	8
Layer no.	3	area =	2	depth =	14
Layer no.	4	area =	4	depth =	19
Concrete properties					
Comp. strength (psi)		beta1	Concrete Crushing Strain		0.003
	4000	0.85			
Conc. strain at max comp. stress (Hognestad's Model)=			0.002		
Slope of stress vs. strain curve beyond peak stress (Hognestad's Model)= z = 150					
Steel properties					
	Yield stress (Ksi)	Yield strain	Elastic Modulus (ksi)		
	60	2.068 E-03	29000		

AXIAL COMPRESSIVE LOAD (P_o)

C_c = 0.85 x 4 x 22 x 22

= 1645.6 K

C_{s1} = 4 x (60 - .85 x 4) = 226.4 K

C_{s2} = 2 x (60 - .85 x 4) = 113.2 K

C_{s3} = 113.2, C_{s4} = 226.4

P_o = 2324.8 K

AXIAL TENSILE LOAD (P_t)

T_{s1} = 4 x 60 = 240 K

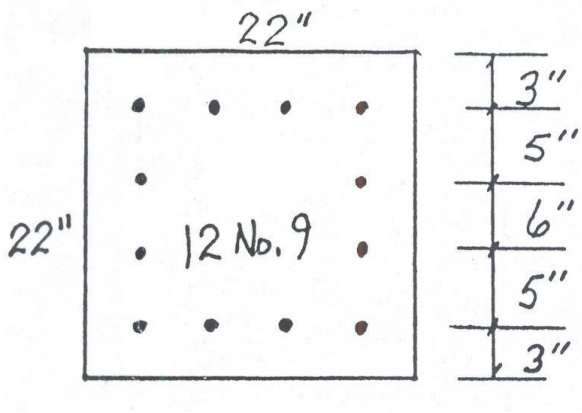
T_{s2} = 2 x 60 = 120 K

T_{s3} = 2 x 60 = 120 K

T_{s4} = 4 x 60 = 240 K

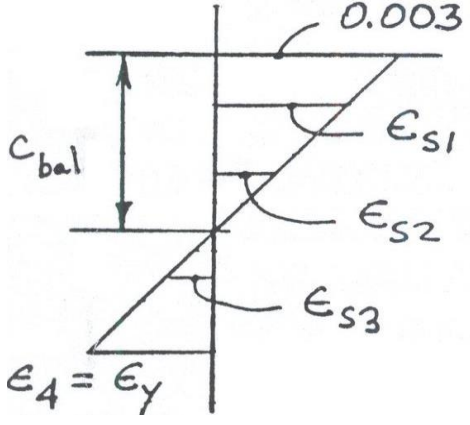
P_t = 720 K

Solved Example: Complete P-M Interaction Diagram



$$f_y = 60\ ksi$$
$$\varepsilon_y = 0.00207$$
$$f_c' = 4000\,psi$$

1: Balanced Point



$$C_{bal} = \frac{0.003}{0.003 + \varepsilon_y}(19'') = 11.2 \text{ in.}$$

$$\varepsilon_{s1} = \frac{c - d_1}{c}(0.003) \quad = 0.00220 > \varepsilon_y$$

$$\varepsilon_{s2} = \frac{c - d_2}{c}(0.003) \quad = 865 \times 10^{-6}$$

$$\varepsilon_{s3} = \frac{c - d_3}{c}(0.003) \quad = -736 \times 10^{-6}$$

$$f_{s1} = 60\ ksi,\ f_{s2} = 25.1\ ksi,\ f_{s3} = 21.3\ ksi$$

$$C_c = 0.85 f_c' b \beta_1 c = 715\ k$$

$$C_{s1} = A_{s1}\left(f_{s1} - 0.85 f_c' \right) = 226\ k$$

$$C_{s2} = A_{s2}\left(f_{s2} - 0.85 f_c' \right) = 43.4\ k$$

$$T_{s3} = A_{s3} \cdot f_{s3} = -42.6\ k$$

$$T_{s4} = A_{s4} \cdot f_{s4} = -240\ k$$

$$P_{bal} = 702\ k$$

Express as Positive

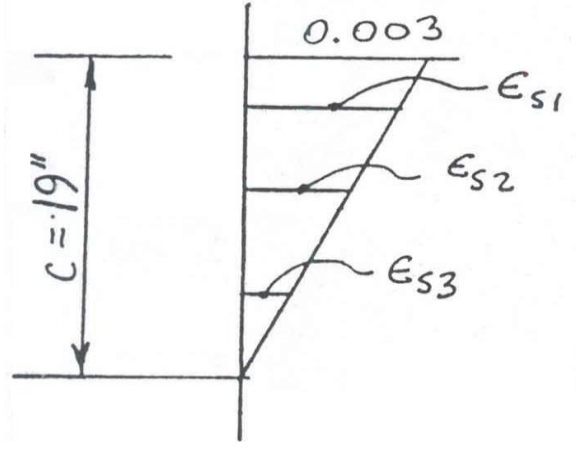
$$M_{bal} = C_c\left(h/2 - \beta_1 c/2 \right) + \sum_i C_{si}\left(h/2 - d_i \right) + \sum_i T_{si}\left(d_i - h/2 \right)$$

$$M_{bal} = 8430 \text{ k-in}$$

$$\Phi_{bal} = 0.003/C_{bal} = 268 \times 10^{-6} \text{ rad/in}$$

2: A Point Above Balanced

Pick $C = d_4 = 19 \text{ in} \Rightarrow \varepsilon_{s4} = 0$



$$\varepsilon_{s1} = \frac{19-3}{19}(0.003) = 0.00253 > \varepsilon_y$$

$$\varepsilon_{s2} = \frac{19-8}{19}(0.003) = 0.00154$$

$$f_{s2} = 50.4 \text{ ksi}$$

$$\varepsilon_{s3} = \frac{19-14}{19}(0.003) = 790\times10^{-6}$$

$$f_{s3} = 22.9 \text{ ksi}$$

$$C_c = 0.85 f_c' b \beta_1 c = 1208\ k$$

$$C_{s1} = A_{s1}\left(f_{s1} - 0.85 f_c'\right) = 226\ k$$

$$C_{s2} = A_{s2}\left(f_{s2} - 0.85 f_c'\right) = 94\ k$$

$$C_{s3} = A_{s3}\left(f_{s3} - 0.85 f_c'\right) = 39\ k$$

$$P = \sum C = 1570\ k$$

$$M = C_c\left(h/2 - \beta_1 c/2\right) + \sum_i C_{si}\left(h/2 - d_i\right)$$

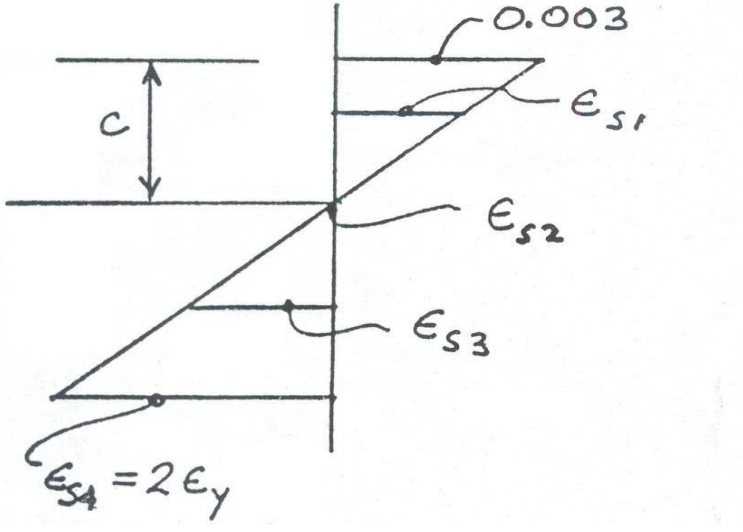
$$= 3533 + 1808 + 282 - 117$$

$$M = 5510 \text{ k-in}$$

$$\Phi = \frac{0.003}{19 \text{ in}} = 158 \times 10^{-6} \text{ rad/in}$$

3: A Point Below Balance, Nominal Diagram

Set $\varepsilon_s = 2\varepsilon_y$



$$C = \left[\frac{0.003}{0.003 + 2\varepsilon_y}\right] d = 7.98''$$

$$\varepsilon_{s1} = 0.00187 \quad , \quad f_{s1} = 54.3 \text{ ksi}$$

$$\varepsilon_{s2} = -6 \times 10^{-6} \quad , \quad f_{s2} = 0$$

$$\varepsilon_{s3} = -0.00226 \quad , \quad f_{s3} = -60 \text{ ksi}$$

$$\varepsilon_{sy} = -2\varepsilon_y \quad , \quad f_{s4} = -60 \text{ ksi}$$

$$C_c = 0.85 f_c' b \beta_1 c = 508\ k$$

$$C_{s1} = A_{s1}\left(f_{s1} - 0.85 f_c' \right) = 204\ k$$

$$T_{s2} = 0$$

$$T_{s3} = A_{s3}\left(- f_y \right) = -120\ k$$

$$T_{s4} = A_{s4}\left(- f_y \right) = -240\ k$$

$$\Big\} \quad P = 352\ k$$

Express as Positive

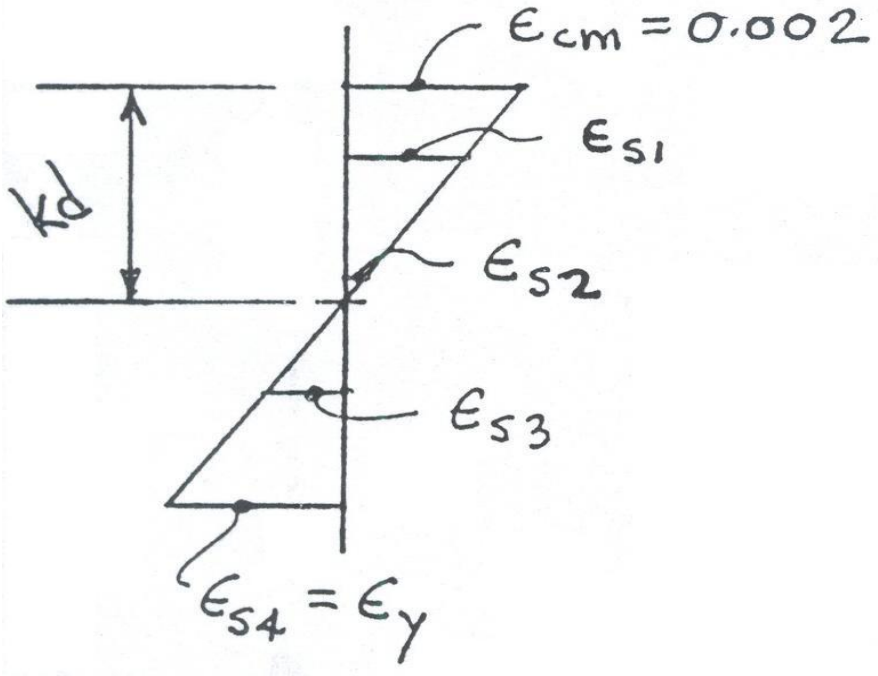
$$M = C_c\left(h/2 - \beta_1 c/2\right) + \sum_i C_{si}\left(h/2 - d_i\right) + \sum_i T_{si}\left(d_i - h/2\right)$$

$$M = 7780 \text{ k-in.}$$

$$\Phi = 0.003/C = 376\times10^{-6} \text{ rad/in.}$$

4 : A Point Below Balance, Yield Diagram

Set $\varepsilon_{cm} = 0.002$ $\quad (k_1 = 0.667 \quad , \quad k_2 = 0.375)$



$$kd = \left[\frac{\varepsilon_{cm}}{\varepsilon_{cm} + \varepsilon_y} \right] d = 9.34 \text{ in.}$$

$$\varepsilon_{s1} = \varepsilon_{c1} = 0.00136$$

$$f_{s1} = 39.4 \text{ ksi} \quad , \quad f_{c1} = 3.59 \text{ ksi}$$

$$\varepsilon_{s2} = \varepsilon_{c2} = 286 \times 10^{-6}$$

$$f_{s2} = 8.30 \text{ ksi} \quad , \quad f_{c2} = 1.06 \text{ ksi}$$

$$\varepsilon_{s3} = -999 \times 10^{-6} \quad , \quad f_{s3} = -29.0 \text{ ksi}$$

$$C_c = k_1 f_c' bkd = 548\ k$$

$$C_{s1} = A_{s1}\left(f_{s1} - f_{c1}\right) = 143\ k$$

$$C_{s2} = A_{s2}\left(f_{s2} - f_{c2}\right) = 14.5\ k$$

$$T_{s3} = A_{s3}\ \ f_{s3} = -58.0\ k$$

$$T_{s4} = A_{s4}\ \ f_{s4} = -240\ k$$

$$\Big\} \quad P = 352\ k$$

Express as Positive

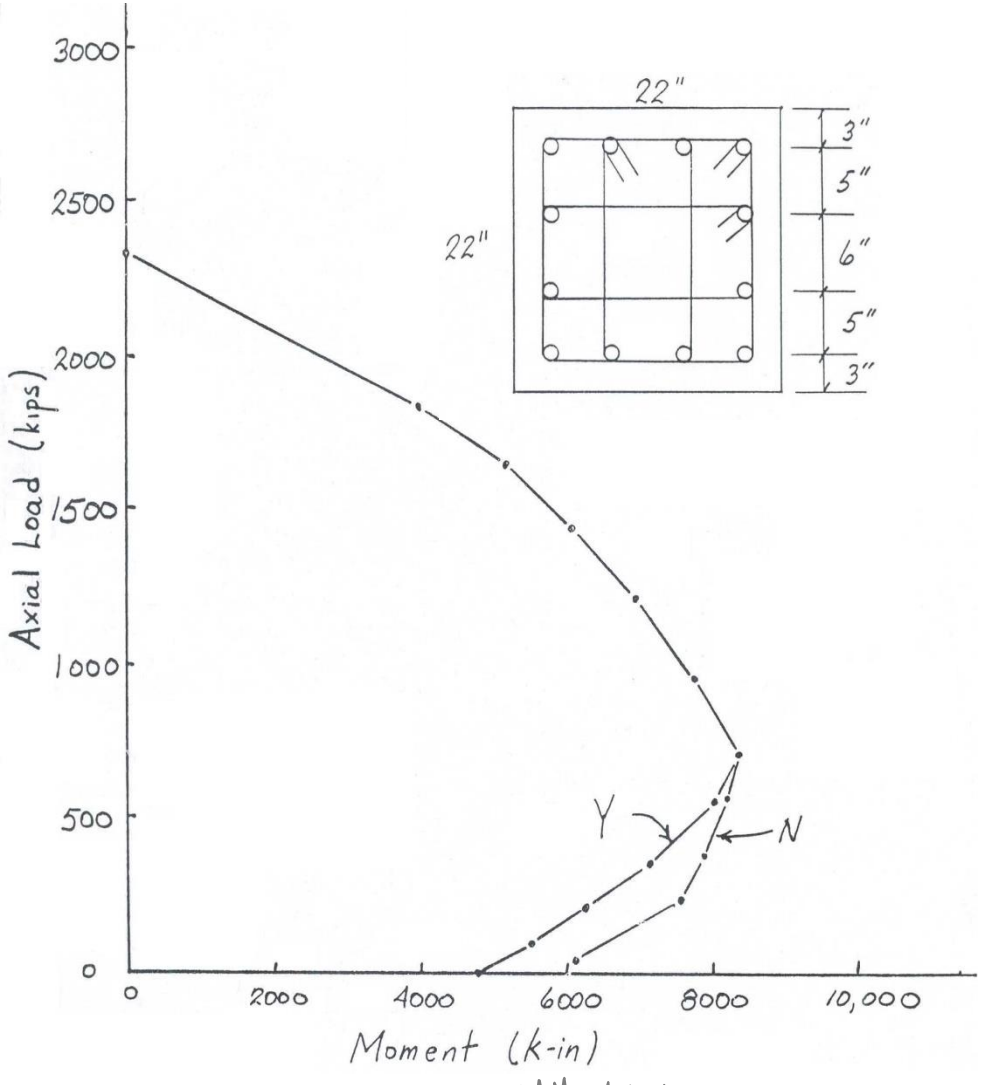
$$M = C_c\left(h/2 - k_2 kd\right) + \sum_i C_{si}\left(h/2 - d_i\right) + \sum_i T_{si}\left(d_i - h/2\right)$$

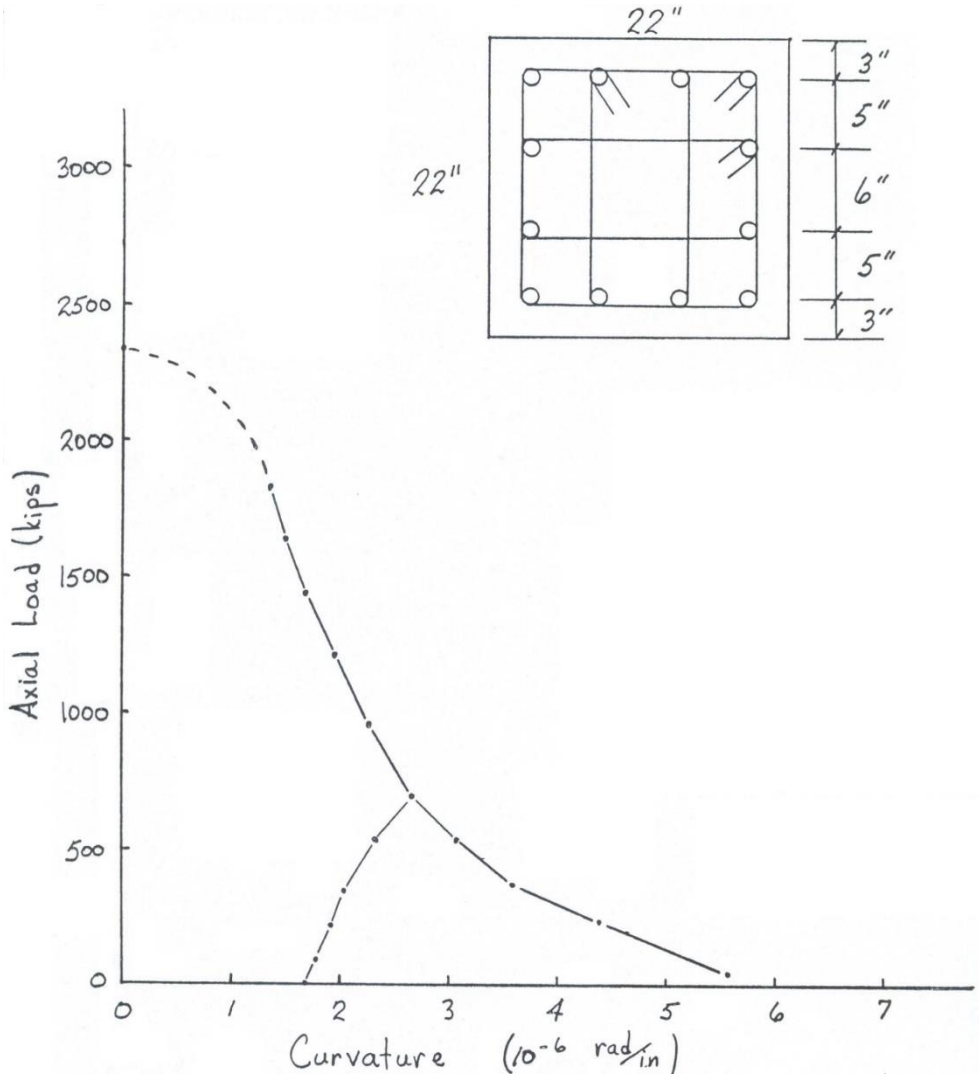
$$M = 7390 \text{ k-in.}$$

$$\Phi = 0.002/kd = 214\times10^{-6} \text{ rad/in.}$$

**** RESULTS FOR NOMINAL DIAGRAM ****			
AXIAL LOAD (KIPS)	NOMINAL MOMENT (K-IN.)	CURAVTURE (RAD./IN.)	
2324.8	0	0	
1819.415	3990.68	1.36364	E-04
1724.501	4630.547	1.43541	E-04
1626.811	5219.912	1.51515	E-04
1539.457	5652.607	1.60428	E-04
1434.626	6149.517	1.70455	E-04
1325.143	6606.003	1.81818	E-04
1210.01	7027.416	1.94805	E-04
1087.924	7420.75	2.0979	E-04
963.9466	7774.937	2.27273	E-04
822.1076	8143.495	2.47934	E-04
702.1297	8436.976	2.66788	E-04
631.755	8374.65	2.854	E-04
557.8183	8286.174	3.06804	E-04
472.0286	8112.867	3.31679	E-04
377.8782	7893	3.60943	E-04
308.777	7572.274	3.95871	E-04
227.7132	7168.089	4.38283	E-04
138.3848	6692.336	4.90873	E-04
37.41104	6127.631	5.57804	E-04
-75.04004	5462.657	6.4587	E-04

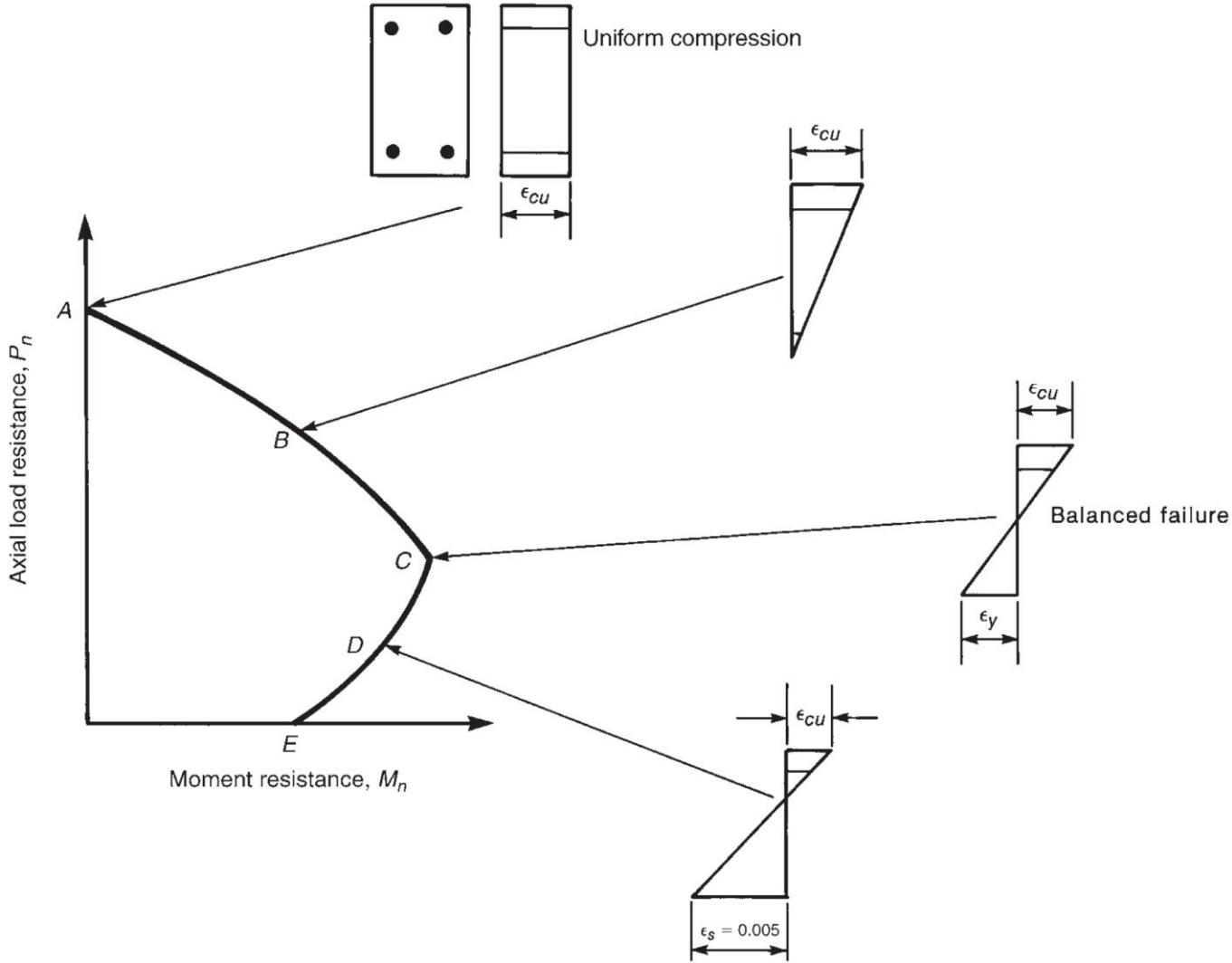
**** RESULTS FOR YIELD DIAGRAM ****			
AXIAL LOAD (KIPS)	MOMENT (K-IN.)	CURAVTURE (RAD./IN.)	
702.1297	8436.976	2.667877	E-04
543.4567	8063.764	2.330024	E-04
362.9939	7160.652	2.091632	E-04
210.9273	6270.086	1.915271	E-04
95.02707	5507.163	1.780181	E-04
-3.005219	4842.844	1.67392	E-04
END OF PROGRAM			


3000
2500
2000
1500
1000
500
0
Axial Load (kips)
0
2000
4000
6000
8000
10,000
Moment (k-in)
Y
N
22"
22"
3"
5"
6"
5"
3"


3000
2500
2000
1500
1000
500
0
Axial Load (kips)
0
1
2
3
4
5
6
7
Curvature (10-6 rad/in)
22"
22"
3"
5"
6"
5"
3"

Properties of P-M Interaction Diagrams of RC Columns

Strain profiles corresponding to points at interaction diagram

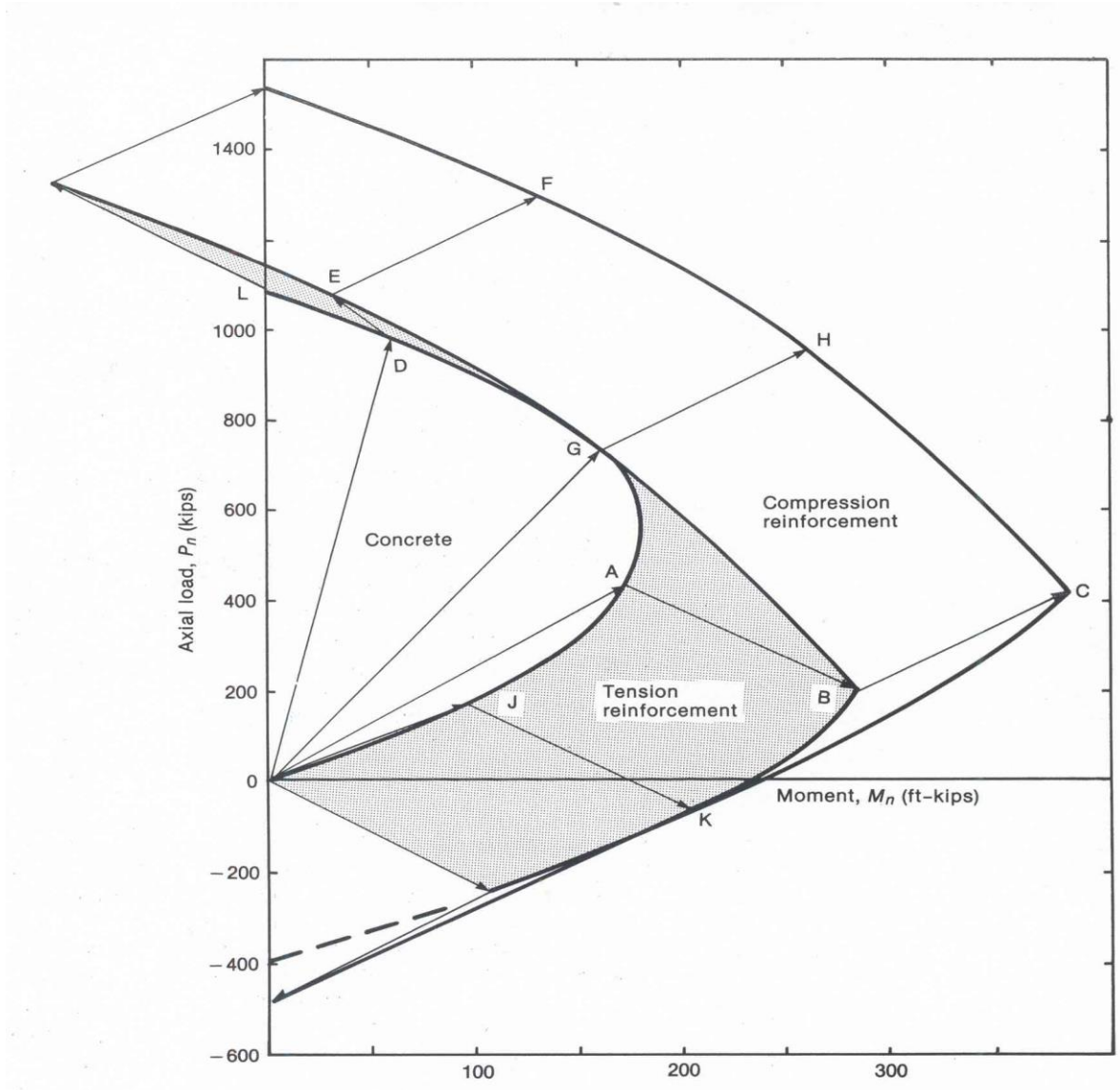


Wight and Macgregor (2012): Reinforced Concrete Mechanics and Design, 6th Edition.

TABLE 11-1 Strain Regimes and Strength-Reduction Factors, ϕ, for Columns and Beams

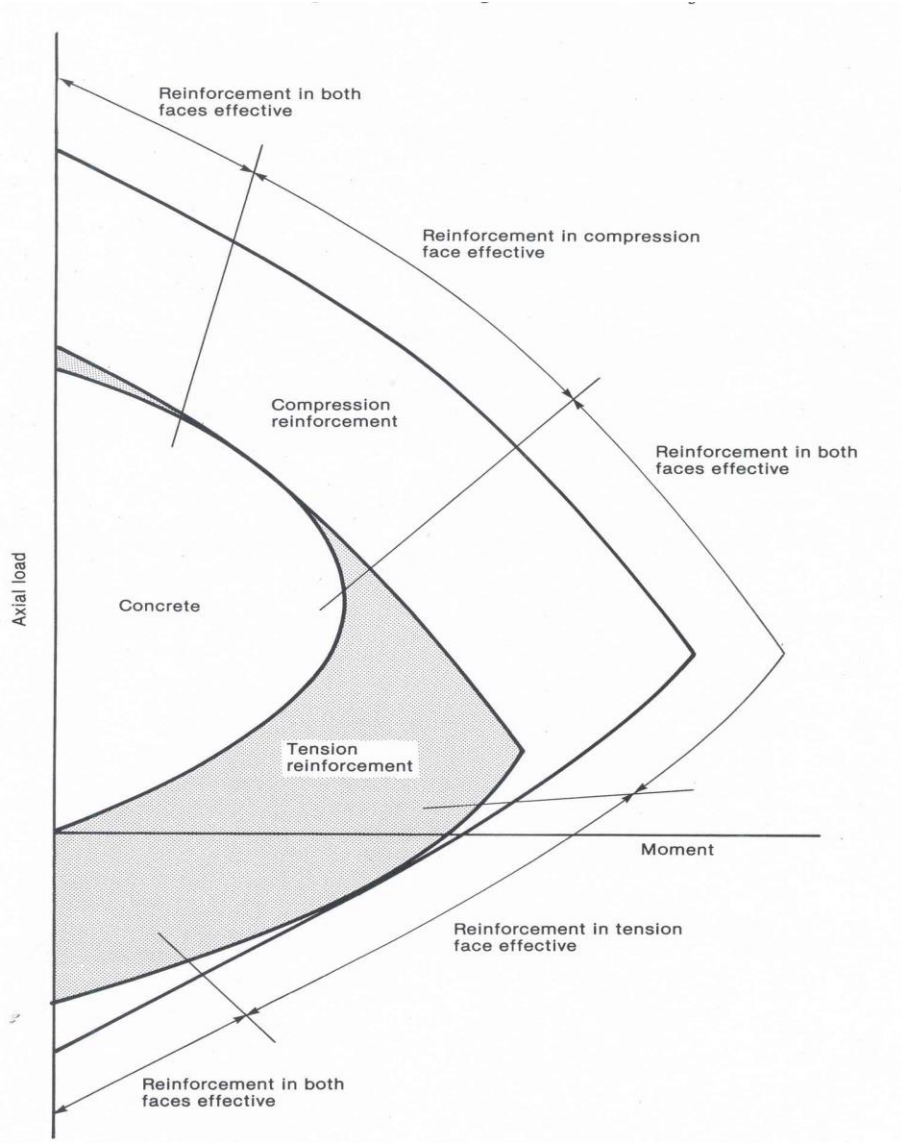
Maximum Strain	Compression-Controlled	Transition Region	Tension-Controlled
Max Compressive Strain	$\varepsilon_{cu} = 0.003$ compression	$\varepsilon_{cu} = 0.003$ compression	$\varepsilon_{cu} = 0.003$ compression
Maximum Tensile strain at ultimate	ε_t between 0.003 compression strain and 0.002 tension strain	ε_t between 0.002 tension strain and 0.005 tension strain	ε_t equal to or greater than 0.005 tension
ASCE 7 Load Factors $U = 1.2D + 1.6L$	**Values of Strength-Reduction Factor, ϕ**	**Values of Strength-Reduction Factor, ϕ**	**Values of Strength-Reduction Factor, ϕ**
ACI Code Section 9.2	*Tied columns* $\phi = 0.65$	*Tied columns* $\phi = 0.65 + (\varepsilon_t - 0.002)\frac{250}{3}$ (11-4)	*Tied columns* $\phi = 0.90$
	Spiral columns $\phi = 0.75$	*Spiral columns* $\phi = 0.75 + (\varepsilon_t - 0.002)\,50$ (11-5)	*Spiral columns* $\phi = 0.90$
	All from ACI Code Sections 9.3.2.1 and 9.3.2.2		

Contributions of Steel and Concrete to Column Capacity



J. G. Macgregor (1997): Reinforced Concrete Mechanics and Design, 3rd Edition.

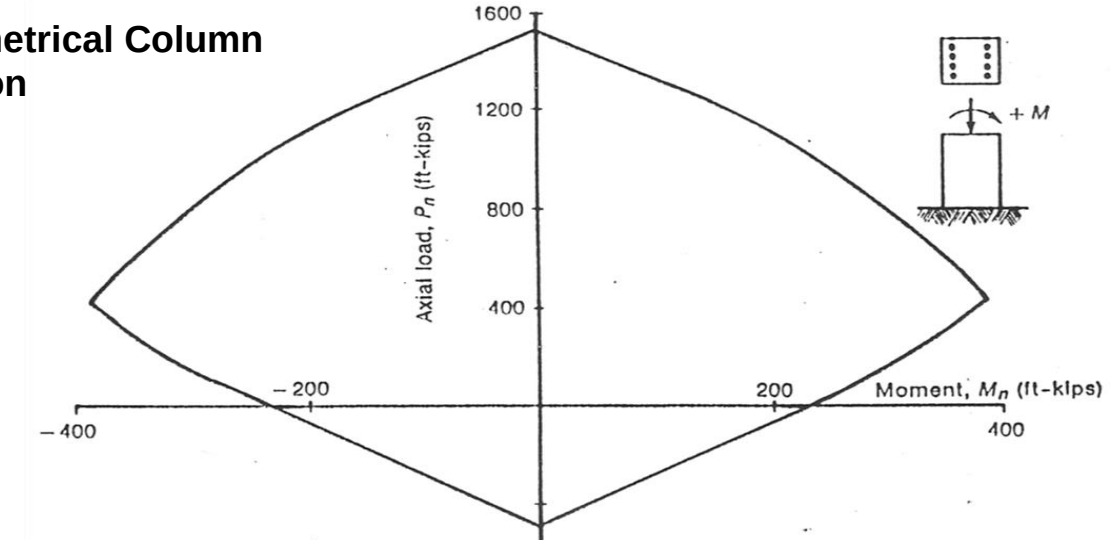
Contributions of Steel and Concrete to Column Capacity



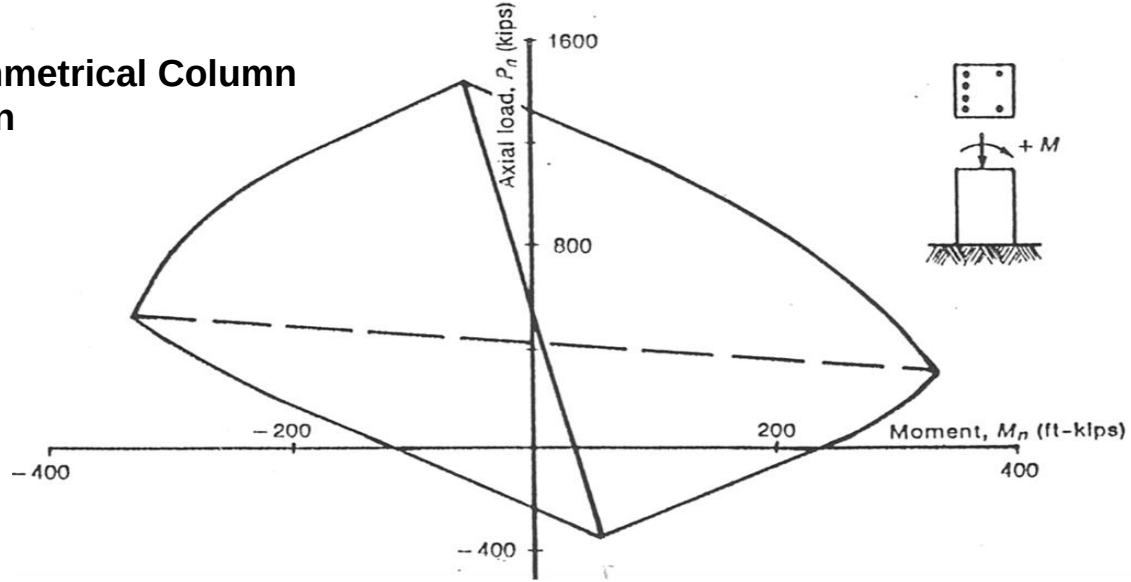
J. G. Macgregor (1997): Reinforced Concrete Mechanics and Design, 3rd Edition.

Effect of Symmetry of RC Column Section

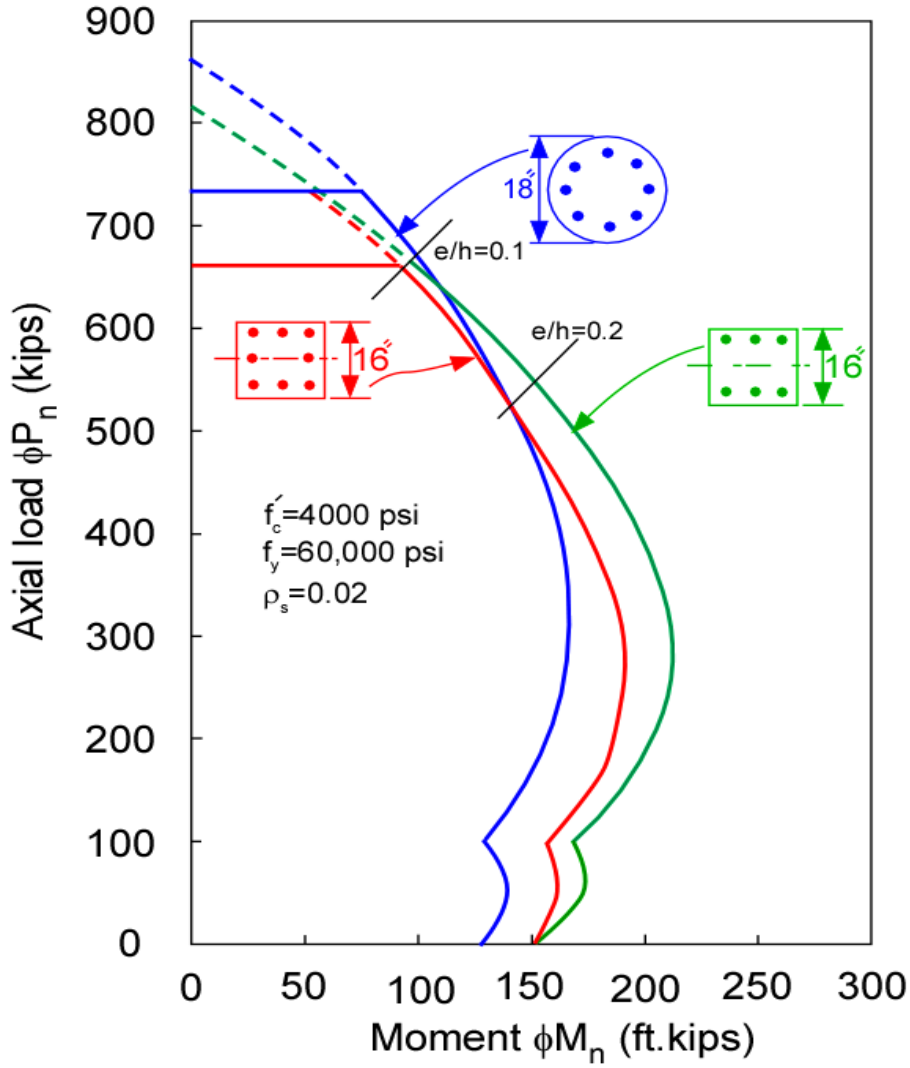
Symmetrical Column Section



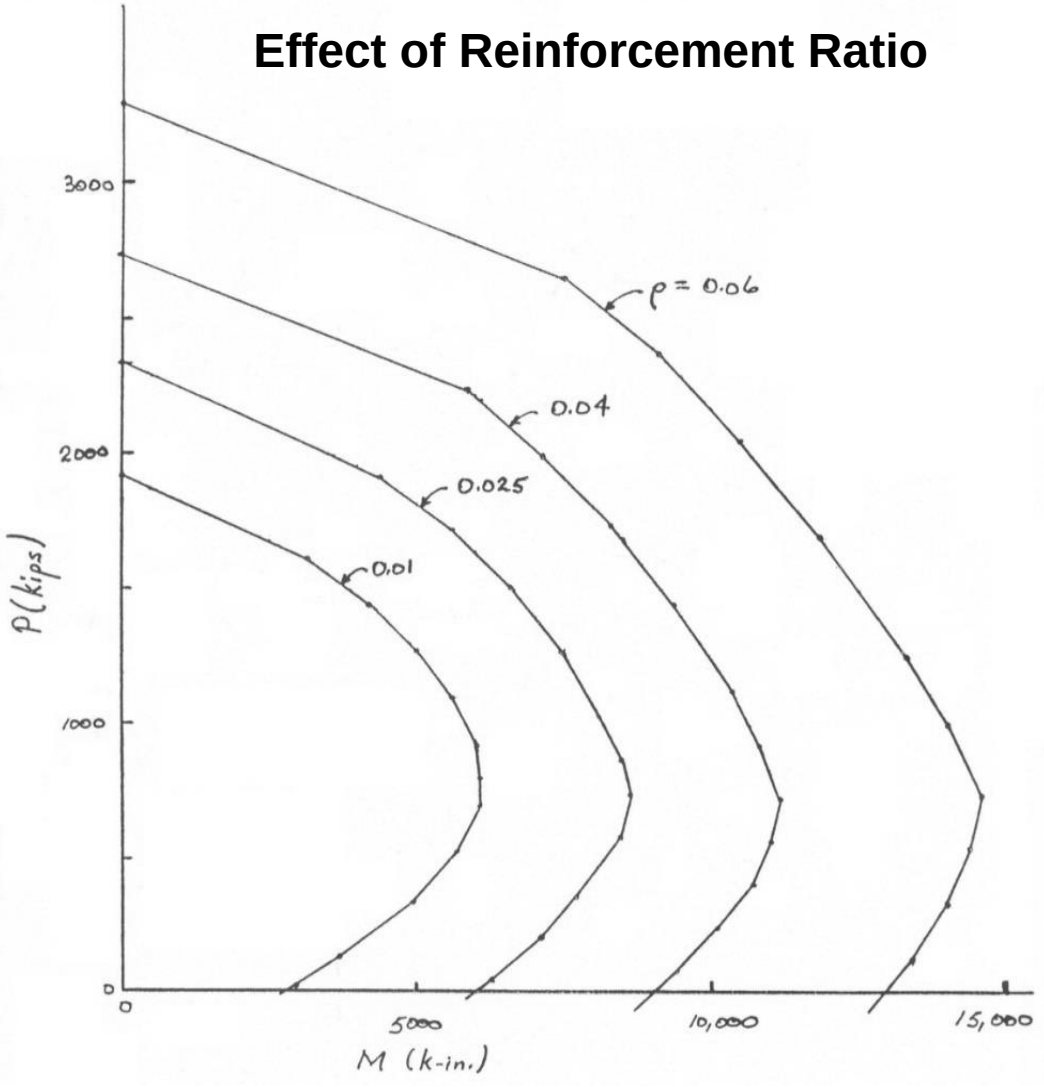
Unsymmetrical Column Section



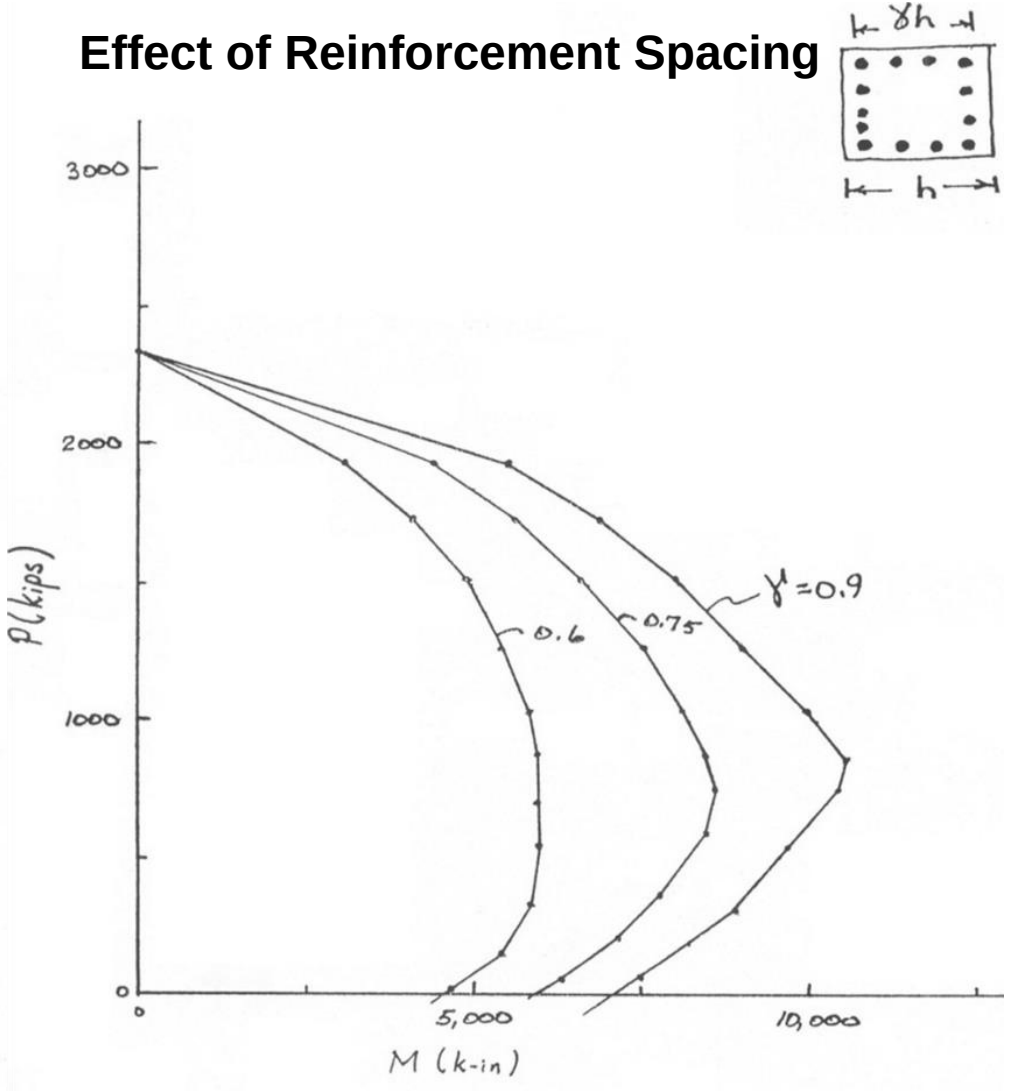
Effect of Column Type on Shape of Interaction Diagram

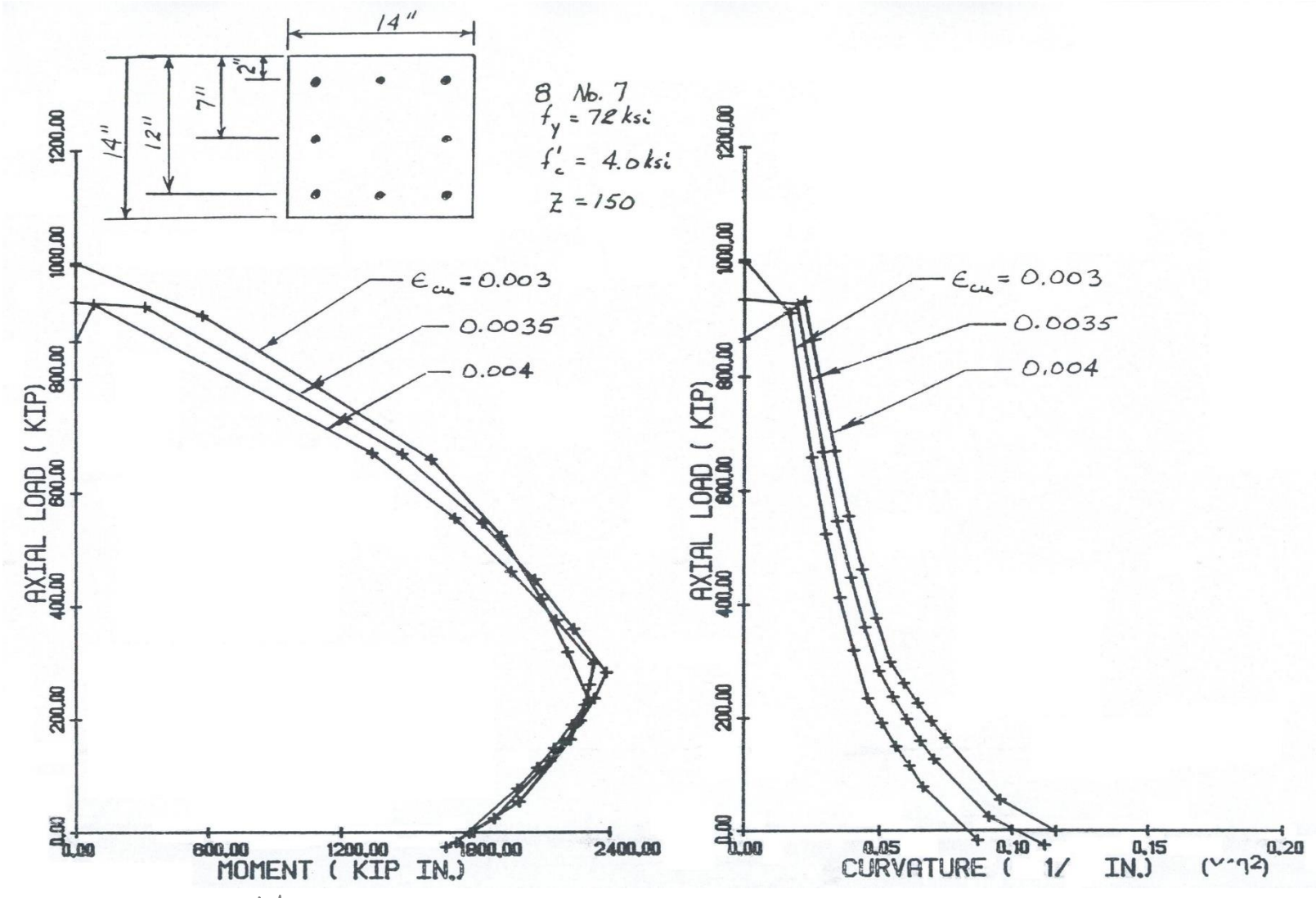


Effect of Reinforcement Ratio



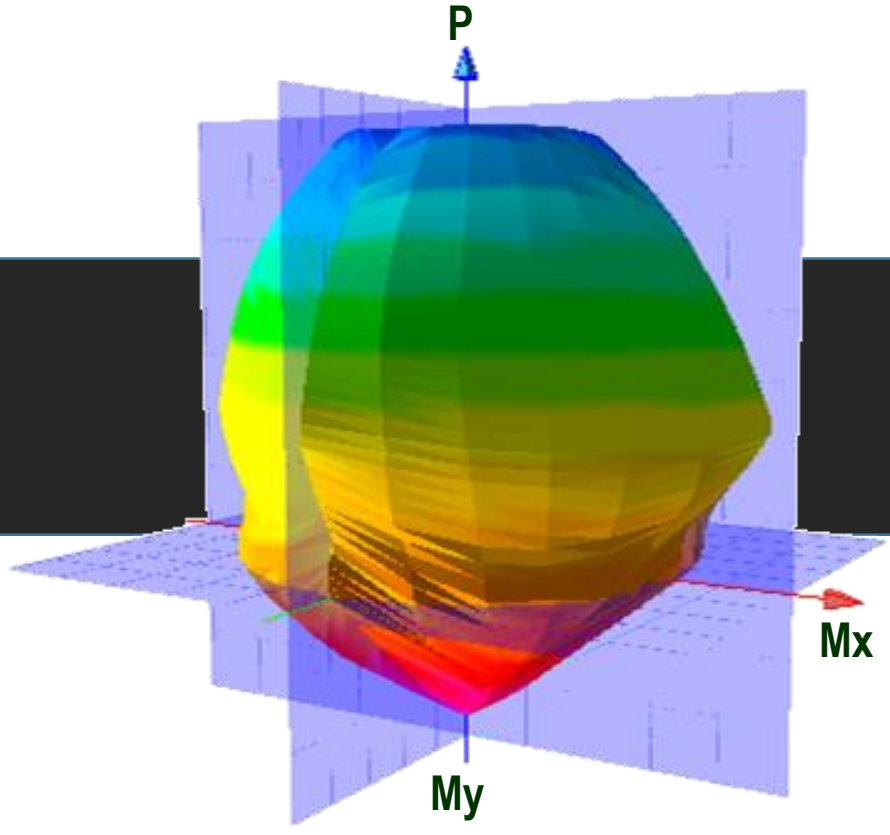
Effect of Reinforcement Spacing




14"
2"
7"
14"
12"
8 No. 7
f_y = 72 ksi
f'_c = 4.0 ksi
Z = 150
ϵ_{cu} = 0.003
0.0035
0.004
AXIAL LOAD (KIP)
MOMENT (KIP IN.)
CURVATURE (1/ IN.)
0.00
200.00
400.00
600.00
800.00
1000.00
1200.00
1800.00
2400.00
0.05
0.10
0.15
0.20

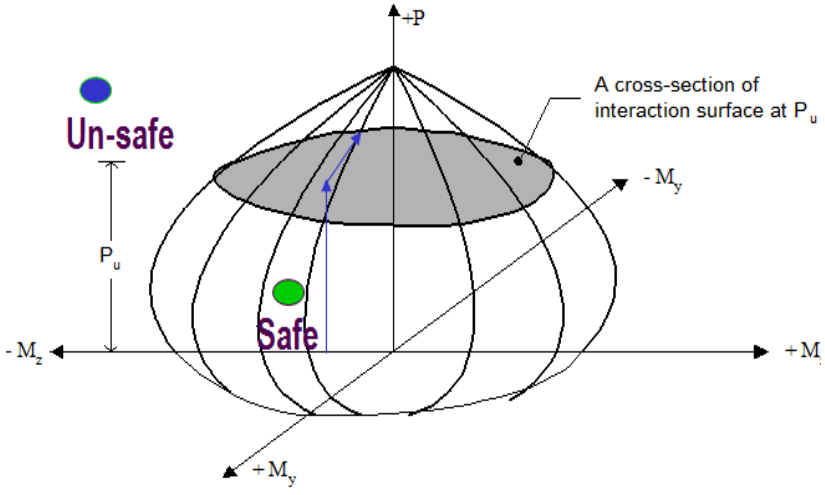
P-M Interaction Surface (Capacity Surface)

Biaxial Bending - Short Columns



$P - M_x - My$ Interaction Surface

- The surface is generated by changing Angle and Depth of Neutral Axis.



$$N_z = \phi_1\left[\frac{1}{\gamma_1}\int_x\int_y \sigma(x,y)dx\,dy \ \ldots + \frac{1}{\gamma_2}\sum_{i=1}^{n} A_i\sigma_i(x,y) \ \ldots\right]$$

$$M_x = \phi_2\left[\frac{1}{\gamma_1}\int_x\int_y \sigma(x,y)dx\,dy\,.y \ \ldots + \frac{1}{\gamma_2}\sum_{i=1}^{n} A_i\sigma_i(x,y)\ y_i \ \ldots\right]$$

$$M_y = \phi_3\left[\frac{1}{\gamma_1}\int_x\int_y \sigma(x,y)dx\,dy\,.x \ \ldots + \frac{1}{\gamma_2}\sum_{i=1}^{n} A_i\sigma_i(x,y)\ x_i \ \ldots\right]$$

$P - M_x - My$ Interaction Surface

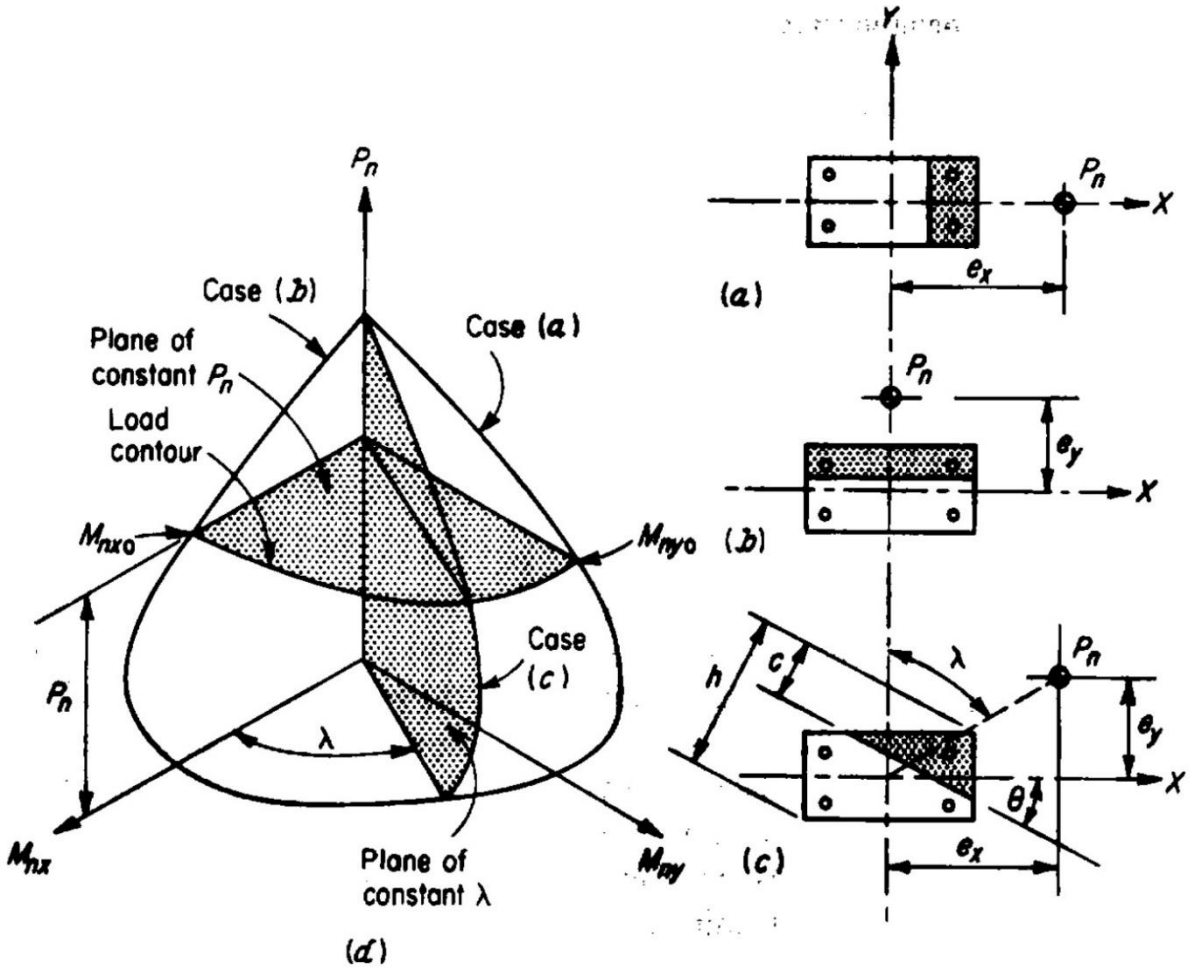
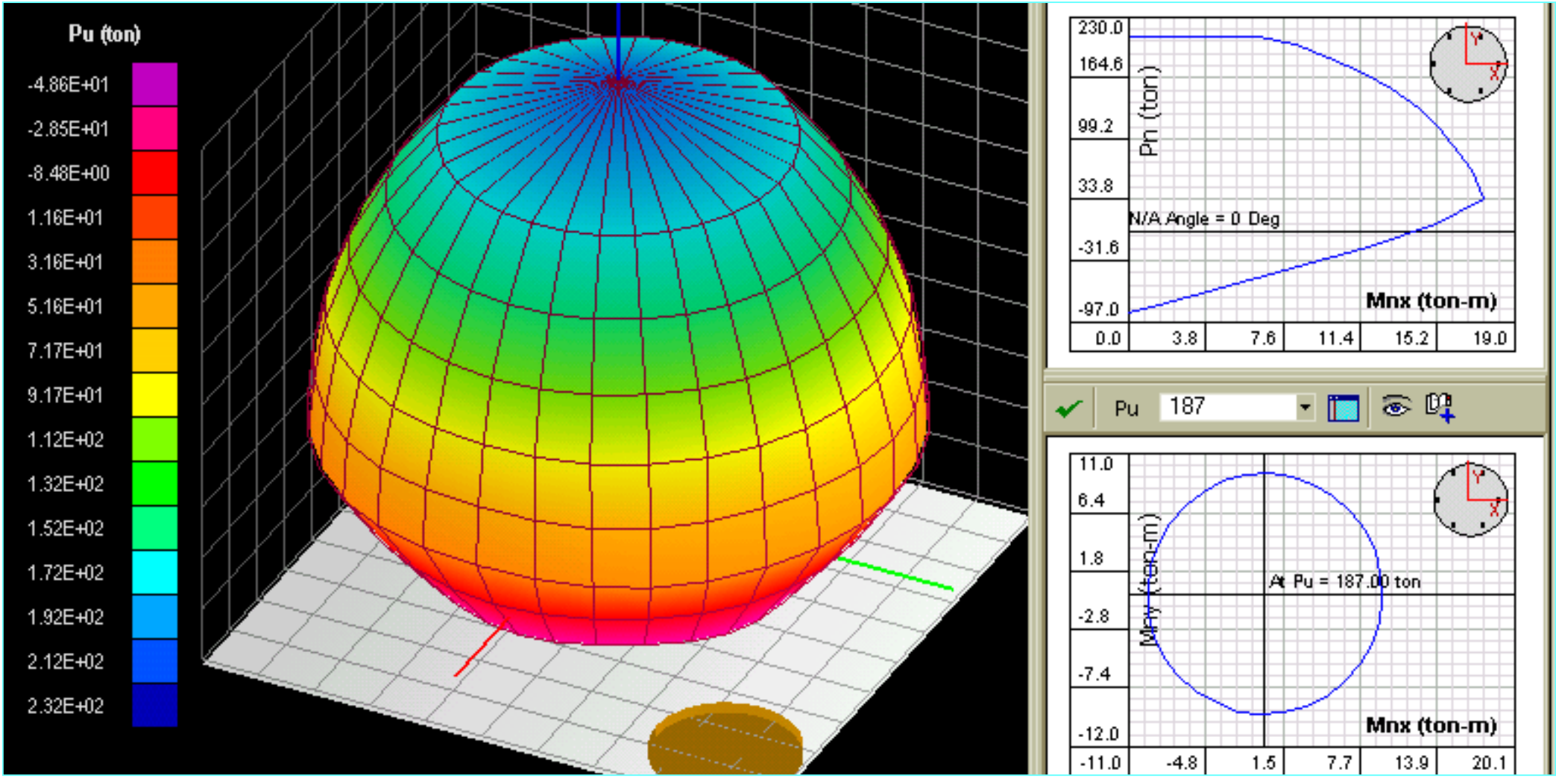
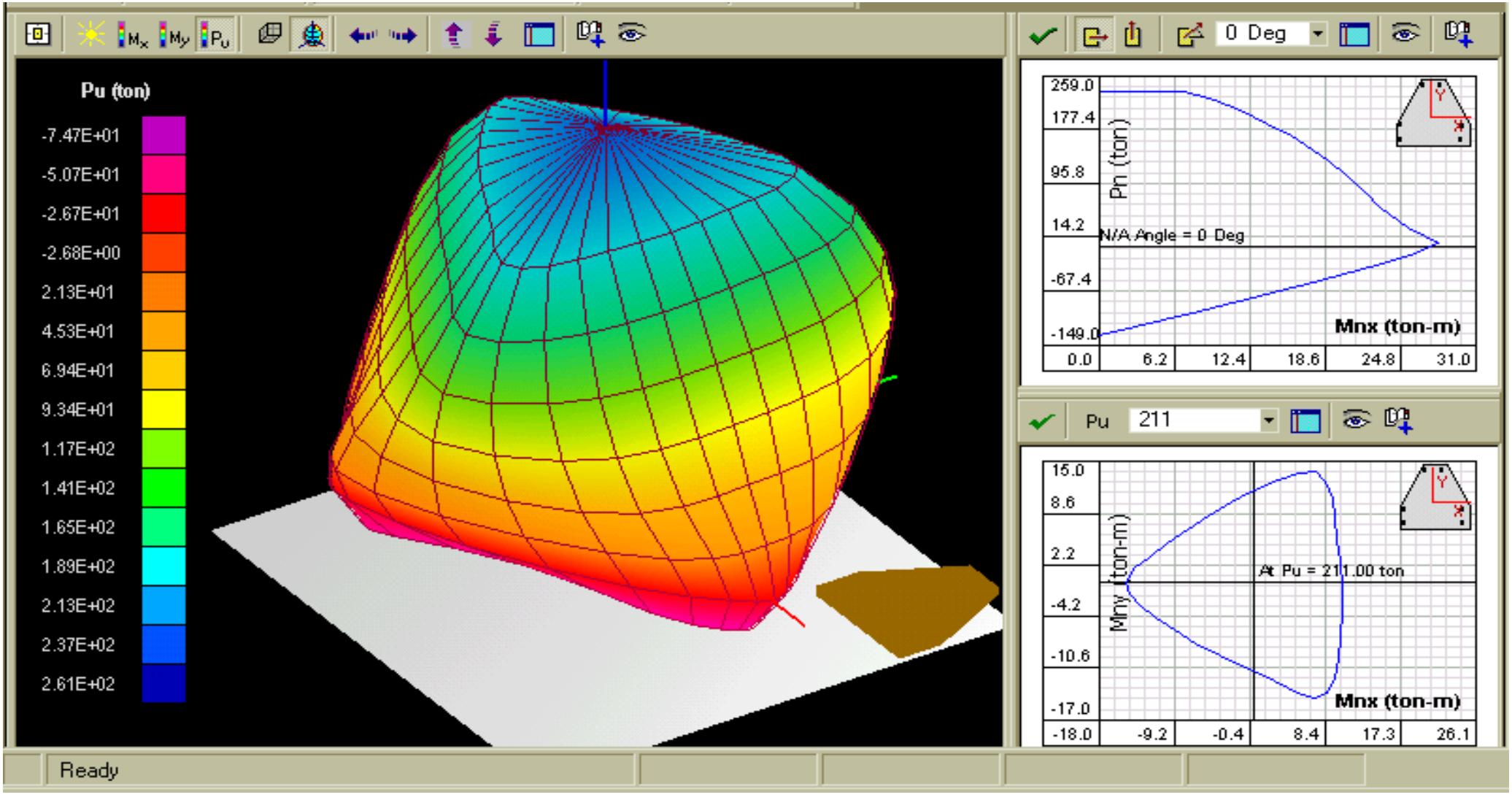
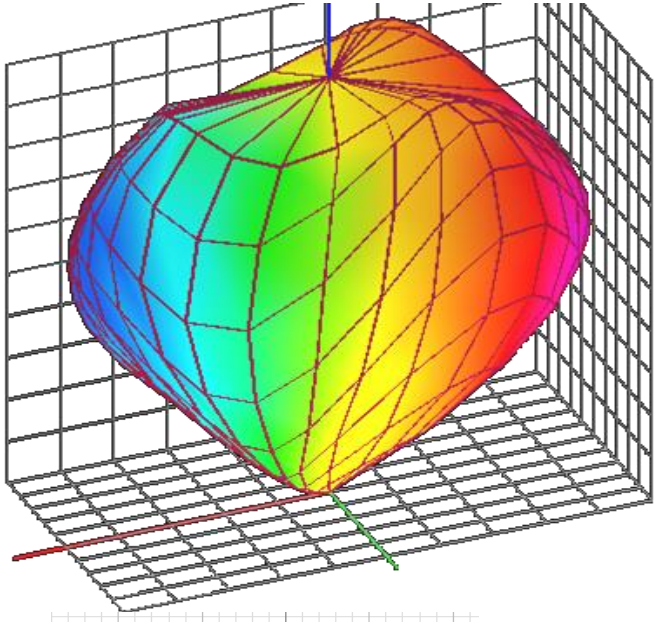
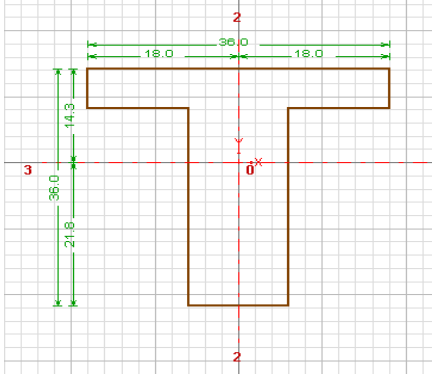


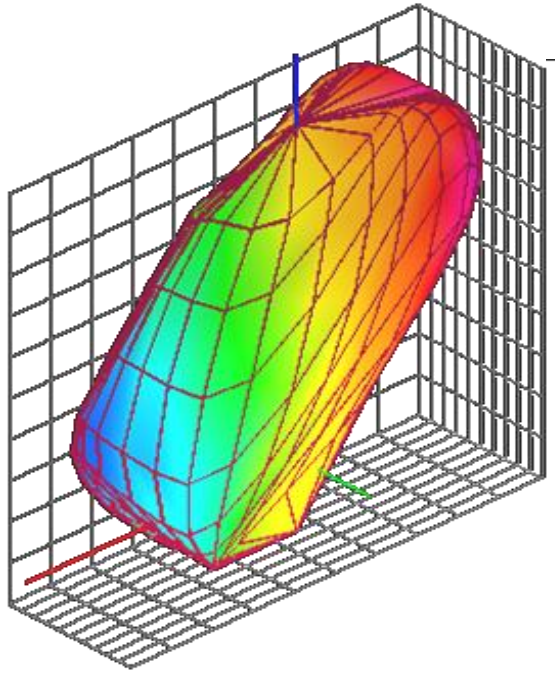
Figure 1. Interaction surface for compression plus biaxial bending: (a) uniaxial bending about Y- axis; (b) uniaxial bending about X-axis; (c) biaxial bending about diagonal axis; and (d) interaction surface.

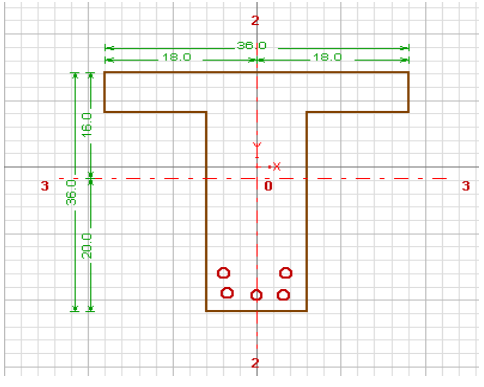

Pu (ton)
-4.86E+01
-2.85E+01
-8.48E+00
1.16E+01
3.16E+01
5.16E+01
7.17E+01
9.17E+01
1.12E+02
1.32E+02
1.52E+02
1.72E+02
1.92E+02
2.12E+02
2.32E+02
230.0
164.6
99.2
33.8
-31.6
-97.0
Pn (ton)
N/A Angle = 0 Deg
Mnx (ton-m)
0.0
3.8
7.6
11.4
15.2
19.0
Pu
187
11.0
6.4
1.8
-2.8
-7.4
-12.0
Mny (ton-m)
At Pu = 187.00 ton
Mnx (ton-m)
-11.0
-4.8
1.5
7.7
13.9
20.1

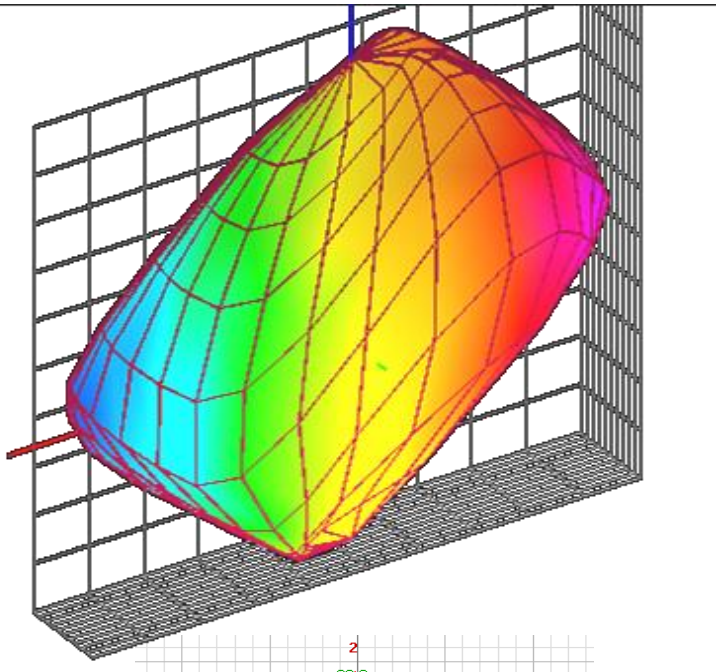

Pu (ton)
-7.47E+01
-5.07E+01
-2.67E+01
-2.68E+00
2.13E+01
4.53E+01
6.94E+01
9.34E+01
1.17E+02
1.41E+02
1.65E+02
1.89E+02
2.13E+02
2.37E+02
2.61E+02
0 Deg
259.0
177.4
95.8
14.2
-67.4
-149.0
Pn (ton)
N/A Angle = 0 Deg
Mnx (ton-m)
0.0
6.2
12.4
18.6
24.8
31.0
Pu
211
15.0
8.6
2.2
-4.2
-10.6
-17.0
Mny (ton-m)
At Pu = 211.00 ton
Mnx (ton-m)
-18.0
-9.2
-0.4
8.4
17.3
26.1
Ready

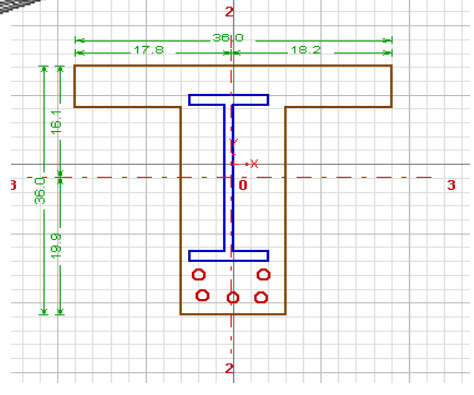



2
36.0
18.0
18.0
14.3
36.0
21.8
3
0
X
Y
2

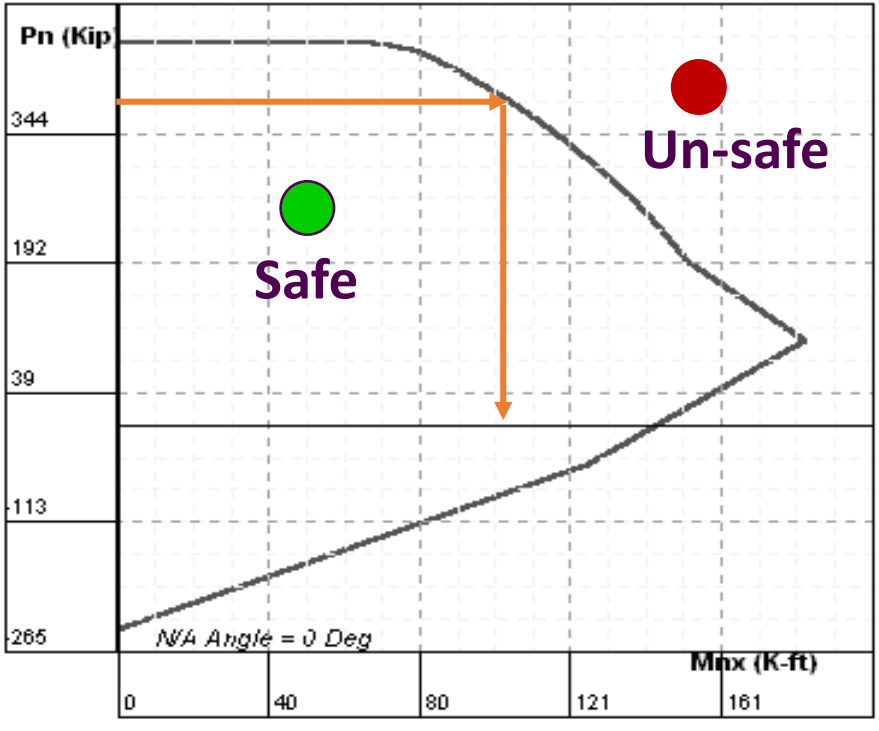



2
36.0
18.0
18.0
16.0
36.0
20.0
3
0
X
Y
2

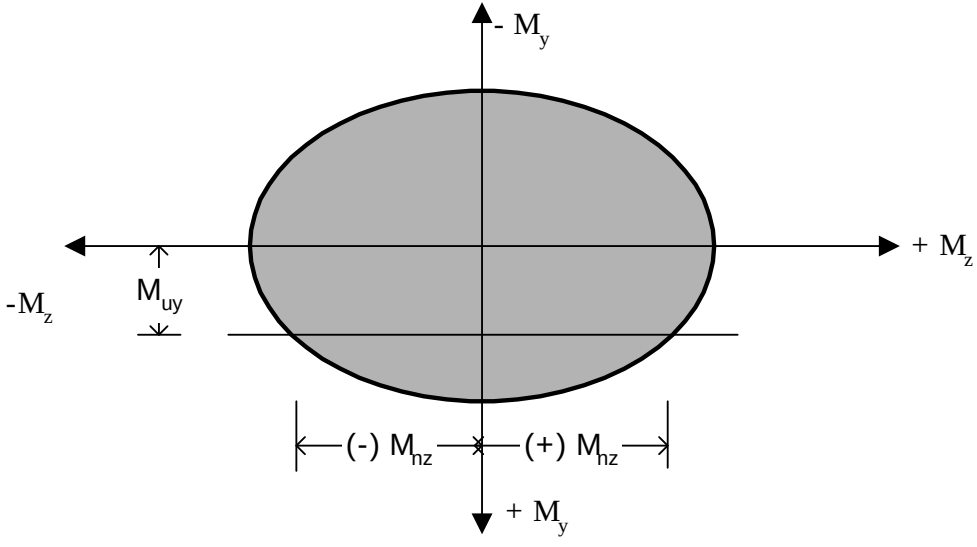



2
36.0
17.8
18.2
16.1
36.0
19.9
3
0
X
Y
2

P-M_x Interaction Curve



M_x-M_y Interaction



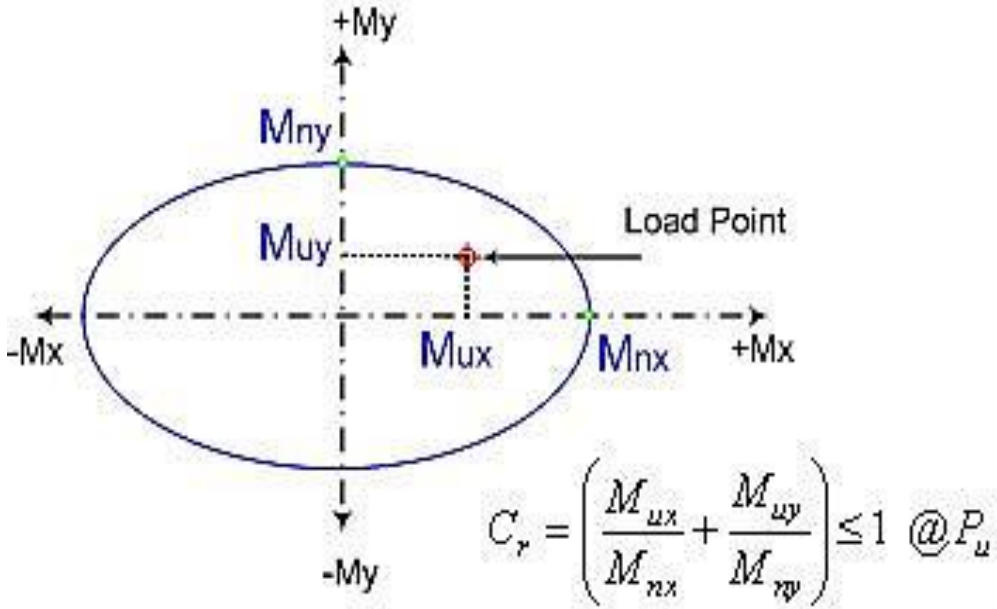
Mx-My Interaction is the basis for many approximate methods

How to Check Capacity?

- How do we check capacity when there are three simultaneous actions and three interaction stress resultants
  - Given: P_u, M_{ux}, M_{uy}
  - Available: P_n-M_{nx}-M_{ny} Surface
- We can use the concept of Capacity Ratio, but which ratio
  - P_u/P_n or M_{ux}/M_{ns} or M_{uy}/M_{ny} or …
- Three methods for computing Capacity Ratio
  - Sum of Moment Ratios at P_u
  - Moment Vector Ratio at P_u
  - P-M vector Ratio

Sum of M_x and M_y

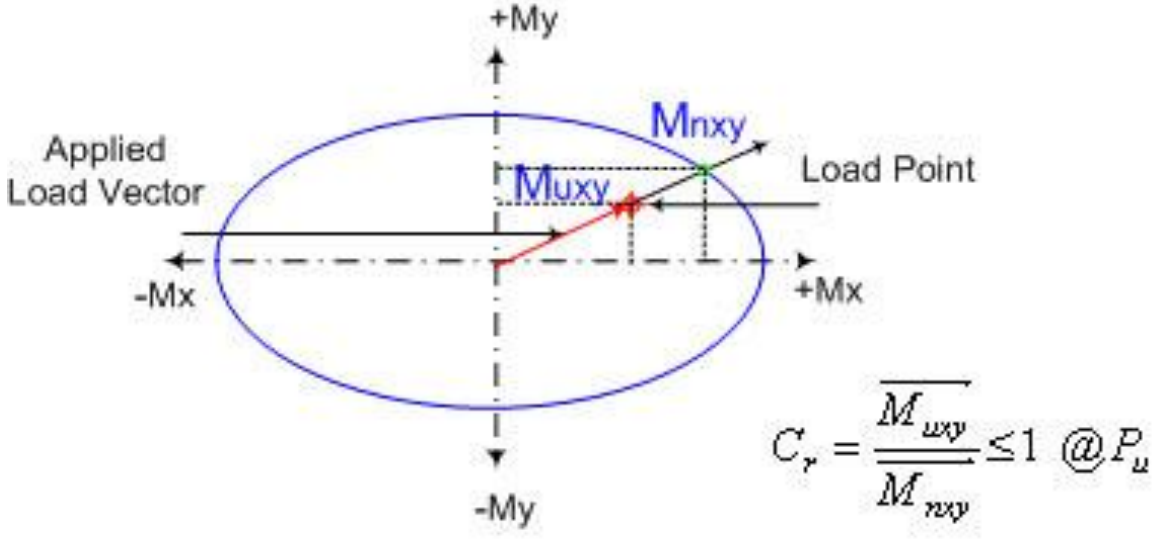
- $M_x - M_y$ curve is plotted at applied axial load, P_u
- Sum of the Ratios of Moment is each direction gives the Capacity Ratio.



Sum of Mx and My Capacities at Pu

Vector Moment Capacity

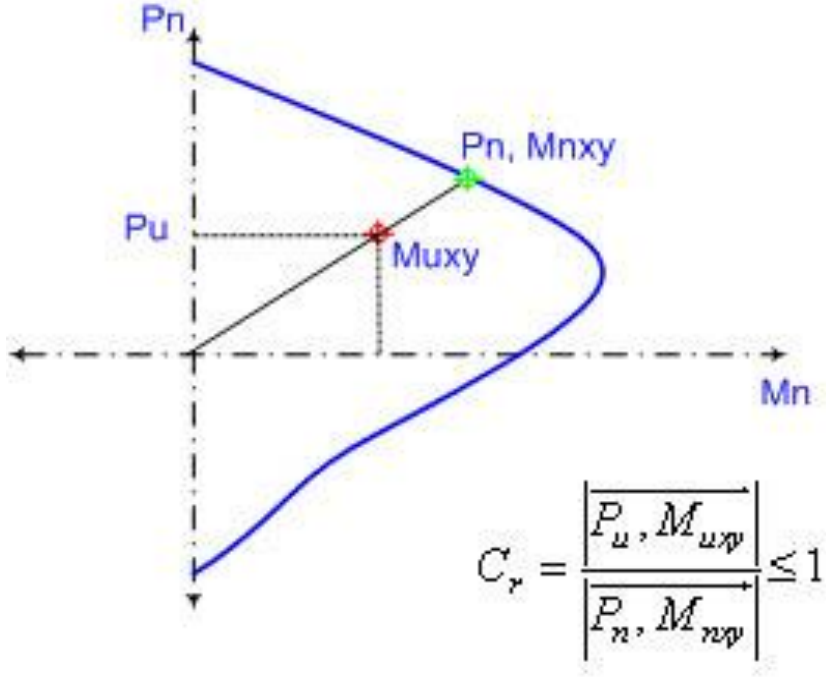
- $M_x - M_y$ curve is plotted at applied axial load.
- Ratio of M_{uxy} vector to M_{nxy} vector gives the Capacity Ratio.



Vector Moment Capacities at Pu

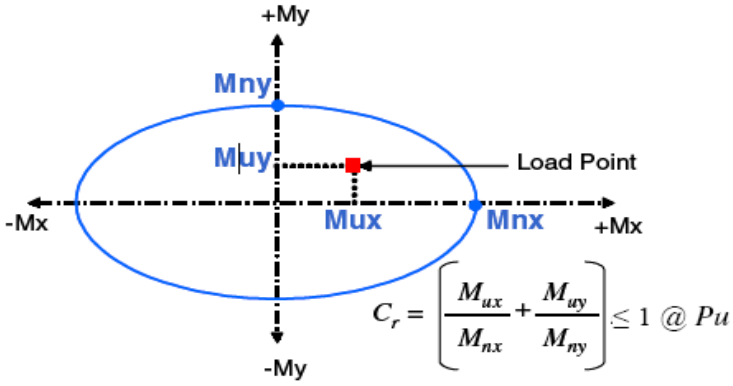
True $P - M$ Vector Capacity

- $P - M$ Curve is plotted in the direction of the resultant moment.
- Ratio of $P_u - M_{uxy}$ vector to $P_n - M_{u_{xy}}$ vector gives the Capacity Ratio.

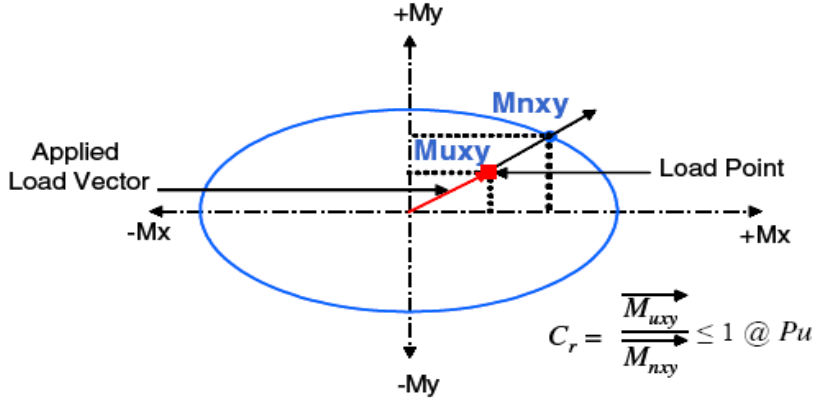


True P-M Vector Capacity

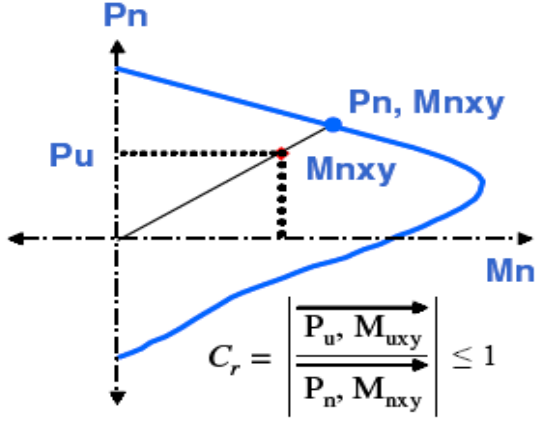
What is Capacity



1- Based on Sum of Moments at Pu

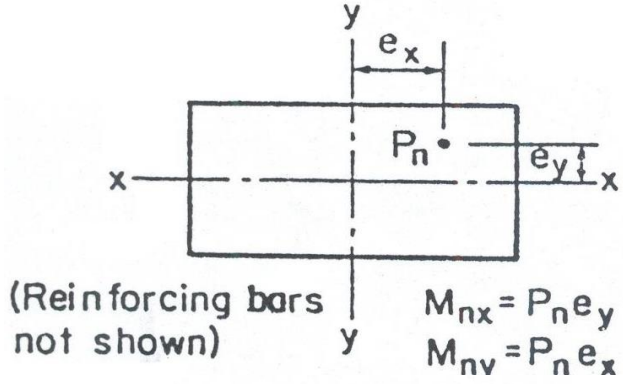


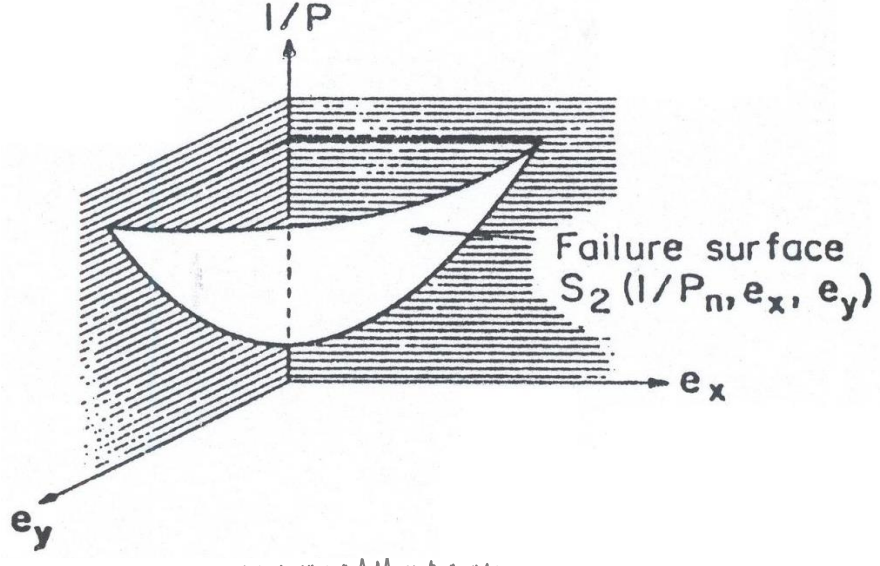
2- Based on Moment Vector at PU

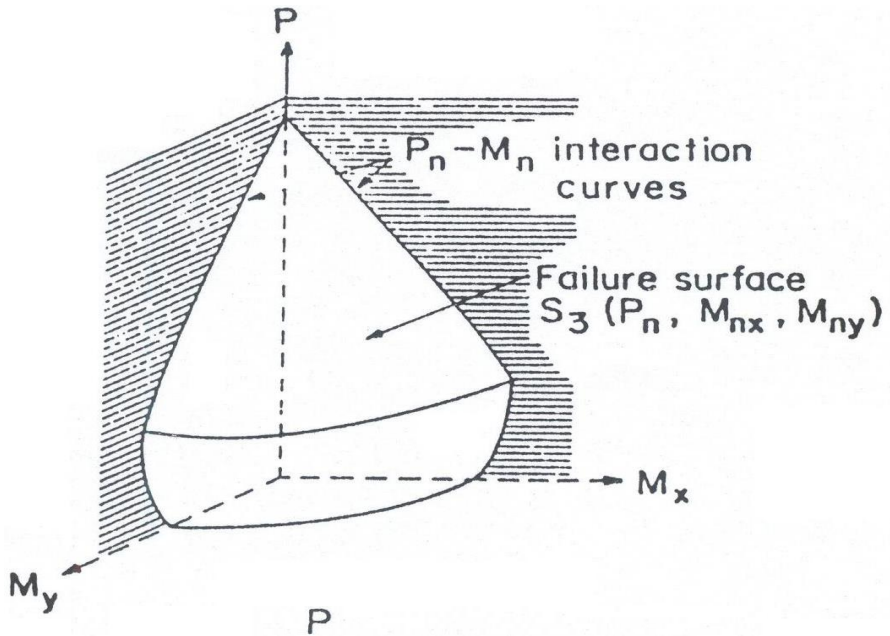


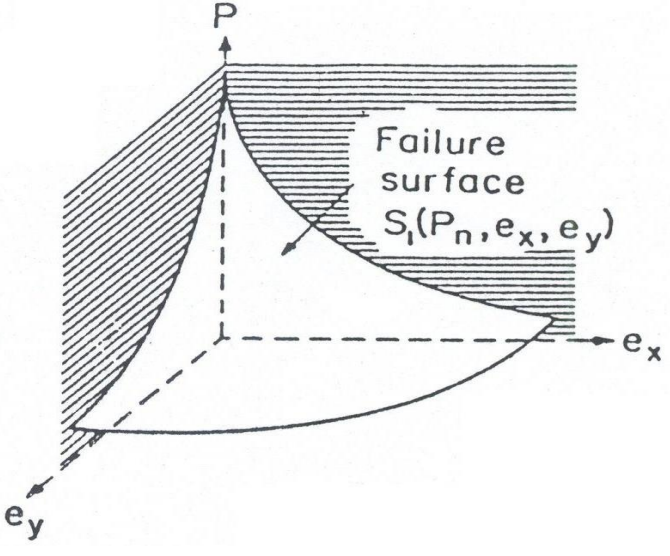
3- Based on True Capacity Vector in 3D

Other Forms of Failure Surfaces









Design of Rectangular Columns Subjected to Biaxial Loads

1) The strain-compatibility method

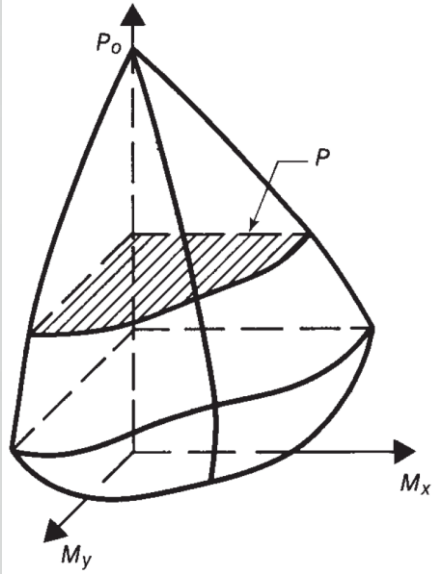
2) The equivalent eccentricity method

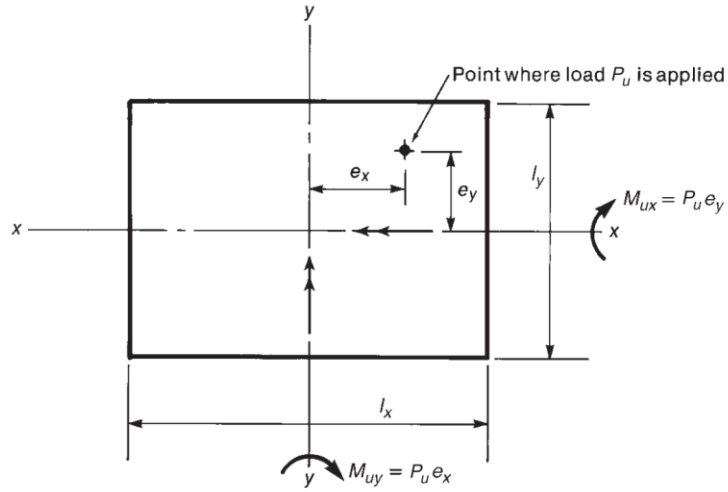
3) Method based on 45-degree slice through interaction surface

4) Load Contour Method (Bresler and Parme)

5) Bresler's reciprocal load method

Design of Rectangular Columns Subjected to Biaxial Loads





P_u = factored axial load, positive in compression.

e_x = eccentricity of applied load measured parallel to the x-axis, positive to right in the cross section in Fig. 11-33.

e_y = eccentricity of applied load measured parallel to the y-axis, positive upward on cross section in Fig. 11-33.

M_{ux} = factoral moment about x-axis, equal to $P_u e_y$, positive when it causes compression in fibers in the positive y direction.

M_{uy} = factoral moment about y-axis, equal to $P_u e_x$, positive when it causes compression in fibers in the positive x direction.

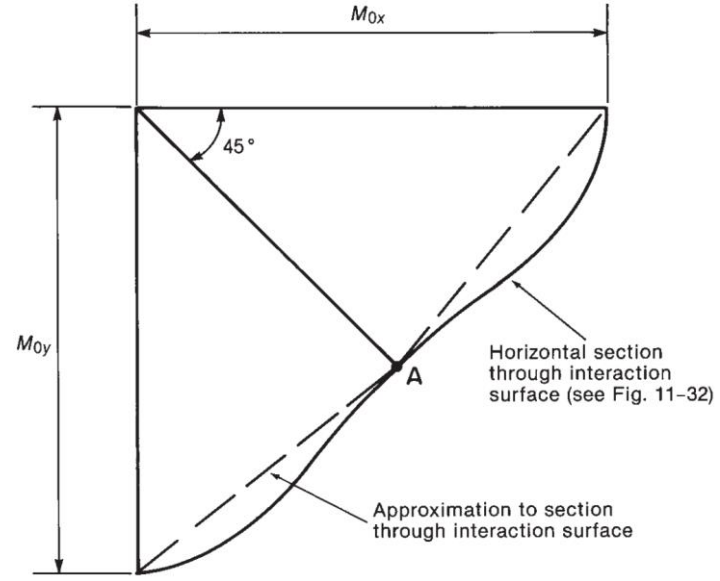
ϕP_{nx} = reduced nominal axial-load capacity for the moment about the x-axis corresponding to the eccentricity, e_y, and the steel and concrete section provided, with $e_x = 0$.

ϕP_{ny} = reduced nominal axial-load capacity, for the moment about the y-axis corresponding to the eccentricity, e_x, and the steel and concrete section provided, with $e_y = 0$.

ϕP_{no} = reduced nominal axial-load capacity for the steel and concrete section provided, with both $e_y = 0$ and $e_x = 0$.

ℓ_x = length of side of column parallel to the x-axis.

ℓ_y = length of side of column parallel to the y-axis.



The equivalent eccentricity method

2. ***The equivalent eccentricity method.*** The biaxial eccentricities, e_x and e_y, can be replaced by an equivalent uniaxial eccentricity, e_{ox}, and the column designed for uniaxial bending and axial load [11-15], [11-16]. We shall define e_x as the component of the eccentricity parallel to the side l_x and the x-axis, as shown in Fig. 11-33, such that the moments, M_{uy}, and M_{ux}, are

$$M_{uy} = P_u e_x \qquad M_{ux} = P_u e_y \tag{11-27a,b}$$

If

$$\frac{e_x}{\ell_x} \geq \frac{e_y}{\ell_y} \tag{11-28}$$

then the column can be designed for P_u and a factored moment $M_{oy} = P_u e_{ox}$, where

$$e_{ox} = e_x + \frac{\alpha e_y \ell_x}{\ell_y} \tag{11-29}$$

where for $P_u/f'_c A_g \leq 0.4$,

$$\alpha = \left(0.5 + \frac{P_u}{f'_c A_g}\right)\frac{f_y + 40{,}000}{100{,}000} \geq 0.6 \tag{11-30a}$$

and for $P_u/f'_c A_g > 0.4$,

$$\alpha = \left(1.3 - \frac{P_u}{f'_c A_g}\right)\frac{f_y + 40{,}000}{100{,}000} \geq 0.5 \tag{11-30b}$$

In Eq. (11-30), f_y is in psi. If the inequality in Eq. (11-28) is not satisfied, the definition of the x and y axes should be interchanged.

Bresler's reciprocal load method

4. ***Bresler reciprocal load method.*** ACI Commentary Sections 10.3.6 and 10.3.7 give the following equation, originally presented by Bresler [11-18], for calculating the capacity under biaxial bending:

$$\frac{1}{\phi P_n} = \frac{1}{\phi P_{nx}} + \frac{1}{\phi P_{ny}} - \frac{1}{\phi P_{no}} \tag{11-31}$$

P_u = factored axial load, positive in compression.

e_x = eccentricity of applied load measured parallel to the x-axis, positive to right in the cross section in Fig. 11-33.

e_y = eccentricity of applied load measured parallel to the y-axis, positive upward on cross section in Fig. 11-33.

M_{ux} = factoral moment about x-axis, equal to $P_u e_y$, positive when it causes compression in fibers in the positive y direction.

M_{uy} = factoral moment about y-axis, equal to $P_u e_x$, positive when it causes compression in fibers in the positive x direction.

ϕP_{nx} = reduced nominal axial-load capacity for the moment about the x-axis corresponding to the eccentricity, e_y, and the steel and concrete section provided, with $e_x = 0$.

ϕP_{ny} = reduced nominal axial-load capacity, for the moment about the y-axis corresponding to the eccentricity, e_x, and the steel and concrete section provided, with $e_y = 0$.

ϕP_{no} = reduced nominal axial-load capacity for the steel and concrete section provided, with both $e_y = 0$ and $e_x = 0$.

ℓ_x = length of side of column parallel to the x-axis.

ℓ_y = length of side of column parallel to the y-axis.

EXAMPLE 11-6 Design of a Biaxially Loaded Column: Equivalent Eccentricity Method

Select a tied column cross section to resist factored loads and moments of $P_u = 250$ kips, $M_{ux} = 55$ kip-ft, and $M_{uy} = 110$ kip-ft. Use $f_y = 60$ ksi and $f'_c = 4$ ksi. The first approximate procedure, based on Eqs. (11-27) to (11-30a), will be used.

1. Select a trial section. Assume that $\rho_g = 0.015$. Use a section with bars in the four faces, because the column is loaded biaxially. Then

$$A_{g(\text{trial})} \geq \frac{P_u}{0.40(f'_c + f_y\rho_g)}$$

$$\geq \frac{250}{0.40(4 + 60 \times 0.015)}$$

$$\geq 128 \text{ in.}^2 \text{ or } 11.3 \text{ in. square}$$

Because this column is subjected to biaxial bending, try a 16-in.-square column with eight No. 8 bars, three in each face.

2. Compute γ.

$$\gamma = \frac{(16 - 2 \times 2.5)}{16} = 0.69$$

3. Compute e_x, e_y, and e_{ox}. From the definition of the moments and eccentricities in Fig. 11-33,

$$e_x = \frac{M_{uy}}{P_u}$$
$$= \frac{110 \times 12}{250} = 5.28 \text{ in.}$$
$$e_y = \frac{M_{ux}}{P_u}$$
$$= \frac{55 \times 12}{250} = 2.64 \text{ in.}$$

By inspection, $e_x/\ell_x \geq e_y/\ell_y$; therefore, use Eq. (11-29) as given. If this were not true, you would transpose the x- and y-axes before using it. In any event, we have

$$\frac{P_u}{f'_c A_g} = \frac{250}{4 \times 256} = 0.244 < 0.4$$

Therefore, use (11-30a) to compute α:

$$\alpha = \left(0.5 + \frac{P_u}{f'_c A_g}\right)\left(\frac{f_y + 40{,}000}{100{,}000}\right) \qquad \text{(11-30a)}$$
$$= (0.5 + 0.244)\left(\frac{60{,}000 + 40{,}000}{100{,}000}\right)$$
$$= 0.744$$

From (11-29),

$$e_{ox} = e_x + \frac{\alpha e_y \ell x}{\ell_y}$$
$$= \left(5.28 + 0.744 \times 2.64 \times \frac{16}{16}\right) = 7.24 \text{ in.}$$

Thus, the equivalent uniaxial moment is

$$M_{oy} = P_u e_{ox}$$
$$= 250 \times 7.24 = 1810 \text{ kip-in.}$$

The column is designed for uniaxial bending for $P_u = 250$ kips and the equivalent moment, $M_{oy} = 1810$ kip-in.

4. Use interaction diagrams to determine ρ_g. Because the column has biaxial bending, we will select a section with bars in four faces. The interaction diagrams are entered with

$$\frac{P_u}{A_g} = \frac{250}{256} = 0.977 \text{ ksi}$$

and

$$\frac{M_{oy}}{A_g h} = \frac{1810}{16^3} = 0.442 \text{ ksi}$$

From Figs. A-9a and A-9b,

$$\text{for } \gamma = 0.60, \rho_g = 0.025$$
$$\text{for } \gamma = 0.75, \rho_g = 0.018$$

By linear interpolation, $\rho_g = 0.021$ for $\gamma = 0.69$.

5. Compute A_{st} and select the reinforcement.

$$A_{st} = \rho_g A_g = 0.021 \times 256 = 5.38 \text{ in.}^2$$

Select eight No. 8 bars, three in each face, $\mathbf{A_{st} = 6.32 \text{ in.}^2}$. Design ties and lap splices as in earlier examples. ■

An Example P-M Interaction Diagram used for the Design of Short RC Columns

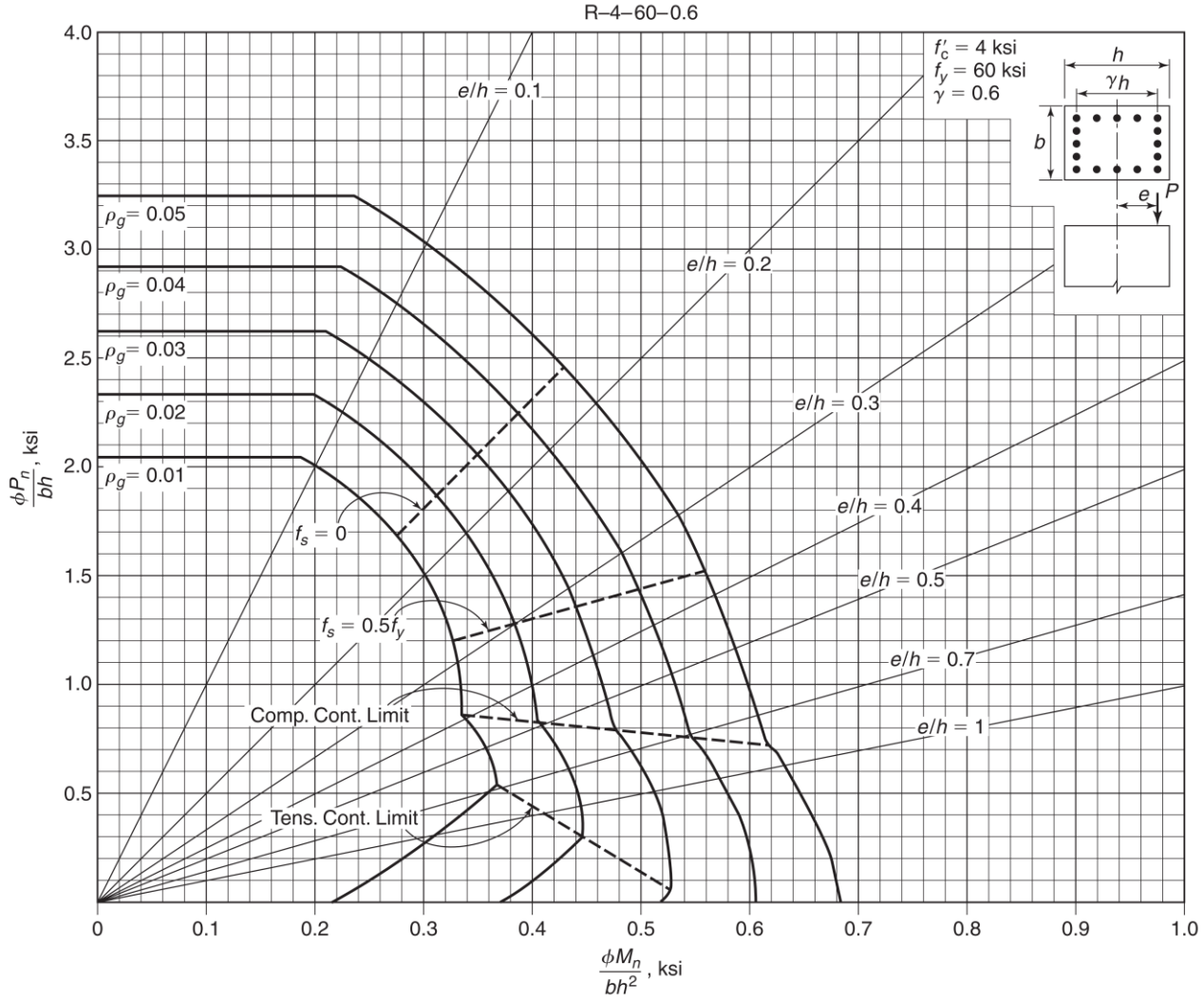


Fig. A-9a
Nondimensional interaction diagram for rectangular tied column with bars in four faces: $f'_c = 4000$ psi and $\gamma = 0.60$.

EXAMPLE 11-7 Design of a Biaxially Loaded Column: Bresler Reciprocal Load Method

Repeat Example 11-6, but use Eq. (11-31).

1. Select a trial section. Select A_g as in Example 11-6. To use Eq. (11-31), it is necessary also to estimate the reinforcement required. Try a 16-in.-square column, $f'_c = 4$ ksi, $f_y = 60$ ksi, eight No. 8 bars (three in each face), and No. 3 ties.

2. Compute γ.

$$\gamma = 0.69$$

3. Compute ϕP_{nx}. ϕP_{nx} is the factored axial load capacity corresponding to e_x and ρ_g. We have

$$\rho_g = \frac{8 \times 0.79}{16 \times 16} = 0.0247$$

$$\frac{e_x}{\ell_x} = \frac{M_{uy}}{P_u \ell_x} = \frac{110 \text{ kip-ft} \times 12}{250 \text{ kips} \times 16 \text{ in.}}$$

$$= 0.330$$

From the interaction diagram, Fig. A-9a, e/h $(e_x/\ell_x) = 0.330$ and $\rho_g = 0.0247$ give $(\phi P_{nx}/bh) = 1.30$ for $\gamma = 0.60$. From Fig. A-9b, $(\phi P_{nx}/bh) = 1.40$ for $\gamma = 0.75$. Interpolating gives $(\phi P_{nx}/bh) = 1.36$ and $\phi P_{nx} = 348$ kips.

4. Compute ϕP_{ny}.

$$\frac{e_y}{\ell_y} = \frac{M_{ux}}{P_u \ell_y} = \frac{55 \times 12}{250 \times 16} = 0.165$$

From Fig. A-9a, $(\phi P_{ny}/bh) = 2.10$ for $\gamma = 0.60$. From Fig. A-9b, $(\phi P_{ny}/bh) = 2.20$ for $\gamma = 0.75$. Interpolating gives $(\phi P_{ny}/bh) = 2.16$ and $\phi P_{ny} = 553$ kips.

5. Compute ϕP_{no}. Use Eq. (11-8b) to calculate $P_{no} = 1230$ kips. Thus, $\phi P_{no} = 0.65 \times 1230 = 798$ kips.

6. Solve for ϕP_n.

$$\frac{1}{\phi P_n} = \frac{1}{\phi P_{nx}} + \frac{1}{\phi P_{ny}} - \frac{1}{\phi P_{no}}$$

$$= \frac{1}{348} + \frac{1}{553} - \frac{1}{798}$$

$$\phi P_n = 292 \text{ kips}$$

The required capacity is 250 kips; therefore, the column design is adequate. **Use eight No. 8 bars, three in each face, $A_{st} = 6.32$ in.^{2}.** ■


Thank You!